

Spacetime–Gauge Commutant Duality: Finite Rigidity and Cofinal Stability

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Abstract

Renewal Geometry reconstructs Lorentzian spacetime and spectral geometry as distinct effective descriptions of a common predictive structure. Here we study their joint finite realization and show that spacetime action and matter multiplicity form an exact double-centralizer pair. The duality is not between the bare spacetime Clifford algebra and the Standard-Model gauge algebra, but between the complete spacetime–gauge–incidence action and the residual multiplicity invisible to that action. This relation admits a positive finite rigidity certificate and remains stable under controlled regulator refinement. On a certified active branch, removing the complete spacetime source leaves an internal structure whose multiplicities reconstruct the Standard-Model nonabelian sectors. A canonical determinant incidence selects $S(U(3) \times U(2)) \cong (SU(3) \times SU(2) \times U(1))/\mathbb{Z}_6$, while anomaly cancellation fixes the hypercharge ratios. A distinct rank-three residual provides a three-generation carrier, and a single grading-odd operator contains the finite Yukawa and Majorana sectors in a common generation frame. The construction is structural rather than numerical: couplings, masses, flavour parameters, renormalization-group flow, and the fermionic measure phase remain independent physical data.

Keywords. Renewal Geometry; Standard Model; spacetime; gauge symmetry; commutant duality; double centralizer; Howe duality; noncommutative geometry; Clifford algebra; generations; hypercharge.

1 Introduction

The Standard Model and general relativity organize physical structure in very different ways. General relativity geometrizes gravity by identifying the metric with the dynamical field of spacetime. The Standard Model instead starts from an internal gauge group, chiral representations, a scalar doublet, and Yukawa couplings, all defined over an already existing spacetime. Under the standard analytic S -matrix hypotheses, the Coleman–Mandula theorem formalizes the familiar separation of continuous spacetime and internal symmetries [3]. A long-standing structural question is nevertheless whether the two sectors, while remaining distinct as physical symmetries, can arise from and constrain one another through a common underlying organization.

Noncommutative geometry gives one of the most developed geometric answers to this question. Early Connes–Lott models already formulated electroweak and Standard-Model sectors as noncommutative gauge theories [4]; in the spectral formulation, geometry is encoded by an algebra, a Hilbert-space representation, and a Dirac operator [5, 6]. Almost-commutative geometries combine an ordinary spin manifold with a finite noncommutative space, and the spectral action recovers the Standard Model coupled to gravity in the corresponding framework [7, 8, 9, 10, 11]. In that construction the continuum spacetime factor and the finite internal factor are specified as parts of the spectral datum. The finite sector itself has a substantial classification theory. Krajewski

diagrams encode finite spectral triples in terms of gauge-multiplet vertices and Yukawa links, and subsequent classifications impose irreducibility and dynamical nondegeneracy [12, 13]. The almost-commutative formulation and its particle-physics interpretation are reviewed systematically in [14, 11]; relaxing spectral-triple conditions can also enlarge the internal symmetry, for example to Pati-Salam type models [15]. Related algebraic reformulations have sought to make the relation among differential forms, spinors, and the finite Standard-Model data more intrinsic [16]. In the conventional Standard-Model finite triple the observed family multiplicity is implemented by taking three copies of the fermionic finite Hilbert space rather than being derived from the spacetime factor [14, 11].

There is also a directly relevant commutant literature inside noncommutative geometry. Morita and Hodge formulations ask when the fermionic Hilbert space is an equivalence bimodule for the spectral Clifford algebra, leading to commutant identities for the Clifford action [17, 18]; in the Standard Model, fermion masses and mixing constrain this Hodge property [19]. These results are close in algebraic form to the duality studied below, but the two statements should not be conflated. Their commutant is built from the left/right spectral Clifford structures of an already specified finite spectral triple. Here the mutually centralizing pair is instead constructed only after the independently reconstructed Lorentzian Clifford action, typed internal action, and physical matter-incidence maps are placed on one predictive carrier; its opposite member is the residual flavour/generation multiplicity algebra. Gauge symmetry of finite noncommutative spaces has likewise been studied intrinsically at the operator-algebraic level [20].

Lorentzian versions of the noncommutative Standard Model provide another important comparison. Barrett formulated a Lorentzian finite geometry for the Standard Model [21], while twisted spectral triples can produce a Krein structure and Lorentzian signature from the spectral data [22]. Outside spectral NCG, Clifford-algebraic approaches have also sought structural relations between spacetime and internal gauge symmetry. Trayling and Baylis derive Standard-Model gauge transformations from a Cl_7 geometric-algebra setting while keeping track of exterior spacetime and interior actions [23]; related Clifford constructions recover the one-generation gauge content in $Cl_{1,5}$ or from minimal ideals of complex Clifford algebras [24, 25]. These are direct precedents for deriving internal structure from an enlarged Clifford carrier, but they do not formulate the support-exact mutual-commutant pair studied here. The present construction differs in logical order: the Lorentzian spacetime packet and the spectral/internal packet are separate companion outputs of one upstream predictive object before their joint commutant is formed.

Renewal Geometry takes a different logical route. Its microscopic entrance is a finite operational process of composable histories and conditional future statistics. Histories that have identical statistics under every admissible future continuation are identified, producing a canonical minimum predictive carrier. Complete positive multitime statistics reconstruct a finite history algebra and its intrinsic positive moment hierarchy, together forming the Renewal Grand Tensor. The companion spacetime work [?] showed that, on a specified minimal faithful branch, calibrated renewal statistics reconstruct a Lorentzian $3 + 1$ ADM geometry and a common-action stationary limit satisfying the vacuum Einstein equation. Independently, the companion spectral work [2] showed that a compatible first-order variation of the reconstructed history algebra produces Connes-type spectral geometry and that spectralization is generally a many-to-one reduction of the underlying predictive dynamics.

Those results leave a natural joint question. Once the Lorentzian and spectral sectors are represented on the *same* source-minimal predictive carrier, what algebraic relation connects spacetime, gauge structure, and matter multiplicity? A tempting answer would be to identify the Standard Model gauge algebra with the commutant of the spacetime Clifford algebra. The first result of the

present paper is that this cannot be correct. A unital copy of $M_4(\mathbb{C})$ acting on a finite Hilbert space has a full matrix algebra as its commutant on each multiplicity block, whereas the typed internal Standard-Model action is reducible. The duality therefore cannot be

$$\text{spacetime Clifford algebra} \longleftrightarrow \text{Standard-Model gauge algebra}.$$

Throughout this paper, “spacetime–gauge duality” therefore refers to a *joint action–multiplicity double-centralizer relation*. It never means that the bare spacetime Clifford algebra and the physical gauge algebra are ordinary commutants of one another.

The correct duality appears one layer later. The external Clifford action, the typed internal action, and the matter-incidence maps generate a *joint typed action algebra*. Its commutant is precisely the residual multiplicity algebra, and finite bicommutant closure gives the converse. In the terminology of invariant theory this is a finite Howe-type situation: two commuting operator algebras determine one another as mutual commutants on the represented carrier [43]. Mutual-centralizer ideas also have direct precedents in quantum field theory. Doplicher–Roberts reconstruction recovers a compact gauge group and field algebra from superselection structure [26, 27], and Todorov exhibited observable–gauge dual pairs in the sense of Howe in four-dimensional conformal field-theory models [28]. Our pair is neither a superselection-category reconstruction nor an observable–gauge dual pair: both members are finite algebras reconstructed on the same source-complete spacetime–matter carrier. We therefore use the term *Howe-type duality* only in this finite mutual-centralizer sense; no oscillator representation or reductive-dual-pair hypothesis is required. The terminology is also unrelated to holographic gauge/gravity duality in the AdS/CFT sense [39, 40].

The first main theorem is therefore

$$\boxed{\mathcal{A}'_{\text{act},X} = \mathcal{M}_{\text{type},X}, \quad \mathcal{M}'_{\text{type},X} = \mathcal{A}_{\text{act},X}.} \quad (1)$$

The statement is support exact. Singular or rectangular matter-incidence maps are treated by their canonical polar supports, positive support metrics, and partial isometries; no inverse Yukawa matrix is introduced. If the support projections fail to commute, the same data generate a canonical operator-space quiver whose endomorphism algebra is the surviving flavour/generation algebra. On the connected invertible branch, all multiplicity spaces can be transported to one root fibre and the dual algebra is generated there by positive polar metrics and cycle holonomies.

A second result makes the duality testable at finite cutoff. If $c_1, \dots, c_s$ generate the action algebra, the commutant Laplacian

$$\mathcal{L}_{\mathcal{C}} = \sum_{j=1}^s \text{ad}_{c_j}^* \text{ad}_{c_j}$$

has kernel exactly the commutant. Restricting this positive form to the orthogonal complement of a proposed protected multiplicity algebra gives a *relative Howe Gram*. Positive definiteness is equivalent to exactness of the proposed multiplicity algebra, while its least eigenvalue is a quantitative rigidity margin. A single positive finite anchor plus a controlled generator tail gives an explicit perturbative stability estimate. Nonperturbatively, transported Mosco convergence together with asymptotic compactness of the commutator forms implies norm-resolvent convergence, convergence of the isolated commutant spectral projection, and cofinal locking of the protected commutant with a positive limiting gap.

The third result identifies an explicitly certified finite Standard-Model realization inside this duality. The tetrahedral spacetime construction carries twelve oriented A_3 roots. The complete spacetime source is removed by its canonical positive Schur short. On the explicit orthogonal active realization this short leaves a faithful twelve-dimensional internal residual, while on a general

common carrier the same residual is measured by the positive shorted Gram $S_{\text{int}}^*(I - P_{\text{ST}})S_{\text{int}}$. The categorical residual decomposes under tetrahedral permutations and root reversal as

$$\mathbb{C}_{\text{type}}^+ \oplus W_+ \oplus (V_2)_+ \oplus W_- \oplus (W \otimes \text{sgn})_- . \quad (2)$$

The direct tetrahedral Burnside blocks are external. The internal factors are controlled instead by the multiplicities of these external irreducibles in the physical word carrier. This gives a finite selection hierarchy: the populated degree-one carrier cannot support an internal $M_3(\mathbb{C})$ factor; a colour factor can first occur only after an external-trivial multiplicity of at least three has appeared; and, once the multiplicity census is $(3, 2, 1, 1)$, one connected colour bridge and one noncommuting weak router are individually source-minimal. A module-recognition theorem distinguishes invariant endomorphisms of a regular word space from operations induced by the historical system algebra. On a common irreducible mediator, reciprocal source routes moreover force a colour bridge through a finite positive return Gram, so the bridge need not be an independent primitive on that route class. Finally, a central-separation test excludes a possible diagonal identification of the two scalar blocks before the full relative commutant is asserted. Under these support-separation hypotheses the four bridge/router deficit dimensions are exactly 9, 11, 13, 15, with the terminal value giving

$$M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C} .$$

Only after this multiplicity-side algebra has landed is the determinant incidence tested. Its nonzero alternating component has stabilizer $S(\text{U}(3) \times \text{U}(2))$. This global form of the Standard-Model group has a long independent history [29, 30]; in ordinary field theory the Lie algebra alone does not experimentally select the full $\mathbb{Z}_6$ quotient, whereas the determinant seed below selects it on the present structural branch. Its central one-parameter subgroup is then fixed, within the displayed tensor-type packet, by anomaly cancellation, yielding the usual hypercharge ratios. This is consistent with the classical anomaly/charge-quantization analysis [31, 32] and with the relation between unimodularity and anomaly cancellation in the noncommutative Standard Model [33].

The reversal-odd standard module $W_- \cong \mathbb{C}^3$ gives the generation carrier; the orthogonal sign-twisted standard module is retained as loop provenance and is not counted as another generation copy. Algebraic explanations of the three-family structure have been pursued by very different mechanisms, including division-algebraic models [35], centralizer characterizations of Standard-Model symmetry [34], and a broad recent survey of algebraic particle-model constraints [36]. Very recent constructions include a Clifford-algebra realization with an intrinsic S_3 family action [37] and an E_7 decomposition into three Standard-Model generation modules [38]. The result here is different: the triplet is not selected from an ambient exceptional algebra or inserted as three copies, but appears as the reversal-odd endpoint source that remains after the complete spacetime source has been shorted on one joint Gram.

The generation statement is therefore not a dimension-count coincidence. A copy of the same three-dimensional tetrahedral representation has already been used by the spacetime branch. The internal copy appears only *after* shorting the complete geometric source on one joint Gram. What survives is a distinct reversal-odd standard source. Its rank is exactly three on the positive branch and is protected by the same finite/cofinal lower-frame tests used in the duality theorem.

Finally, we show that the duality is compatible with the common-action construction of the spacetime paper. If S_g and S_i are the gravitational and internal variation syntheses on one history, with Grams G_g and G_i and mixed block C , the genuinely new internal variation is measured by the Schur quotient

$$\boxed{G_{i|g} = G_i - CG_g^{-1}C^*} . \quad (3)$$

Its rank is the exact additional internal source cost after gravity has been accounted for. On this common carrier the usual finite gauge, Higgs, Dirac–Yukawa, Ward, BRST, and stress identities are derivatives of one action. This finite compatibility is the only dynamical input needed here; the analytic construction of a full Einstein–Standard-Model continuum is left to a separate work.

The scope is deliberately structural. The paper reconstructs the finite internal algebra, physical quotient gauge group, anomaly-fixed hypercharge ratios, a rank-three generation carrier, one primitive weak doublet, and a single finite Dirac/Yukawa operator on the selected finite branch. It does not determine the numerical Yukawa matrices, CKM or PMNS data, gauge couplings, Higgs parameters, renormalization-group running, matching scales, or the determinant/Pfaffian phase of the quantum fermion measure. Nor does the finite commutant theorem alone prove an interacting four-dimensional quantum continuum limit.

Logical dependence on the companion constructions. The results below separate the abstract finite mathematics from its Renewal-Geometric realization. The finite commutant duality, its reciprocal Wedderburn, support-polar and quiver descriptions, the positive finite certificates, and the structural Standard-Model classification are proved from the explicit finite hypotheses stated in this paper. Their proofs do not require the reader to reconstruct the Renewal Grand Tensor, the ADM root-race calibration, or the spectral Hodge calculus. The companion works [?, 2] supply the independently reconstructed Lorentzian and spectral ingredients. The system-bearing mixed incidence and provenance data required by the structural branch are treated by finite occurrence tests. In particular, when the grading, type and incidence compressions are word-native operations of the historical flat algebra, their mixed ancestry Grams are already finite inserted moments of the Grand Tensor; no independent cross-Hankel measurement is then required. The remaining provenance question is whether those typing operations themselves belong to the historical algebra. We also show why coarse record/environment data, or an external shadow not represented in that algebra, cannot determine the acted-on system structure. The final common-action subsection is only a compatibility statement and does not attempt the independent analytic construction of an Einstein–Standard-Model continuum. Accordingly, the finite results may equally be read as theorems about any joint packet satisfying Definitions 2.1 and 2.2.

The conceptual structure of the paper can be summarized as

$$\begin{array}{l}
 \text{predictive Renewal Geometry} \longrightarrow \text{Lorentzian} + \text{spectral joint carrier} \\
 \longrightarrow \text{typed action algebra} \\
 \xleftrightarrow{\text{commutant}} \text{matter multiplicity} \\
 \longrightarrow \text{structural Standard Model.}
 \end{array} \tag{4}$$

Section 2 states the finite joint-carrier inputs axiomatically, records how the two companion constructions realize the relevant external and spectral ingredients, and proves the factor-commutant obstruction that rules out the naive identification of the Standard-Model algebra with the full Clifford commutant. Section 3 proves the exact finite duality in its support-polar, multiplicity-quiver, and metric–holonomy forms. Section 4 reconstructs the structural Standard-Model branch through a multiplicity-birth and router/bridge hierarchy, adds source-native reciprocal-return and central-separation criteria, proves generic exactness away from a Howe enhancement discriminant, identifies source-minimal incidence skeletons, gives finite and cofinal positive certificates for the duality, and records only the source-minimal common-action compatibility needed for later gravity–matter dynamics. Section 5 discusses interpretation and limitations. Additional representation-theoretic and stability details are collected in the appendices.

2 Axiomatic joint carrier and the commutant obstruction

2.1 Abstract Lorentzian Clifford input and its Renewal realization

The spacetime construction of Ref. [?] supplies one Renewal-Geometric realization of the finite external data used here. At finite cutoff X , it first reconstructs the minimum persistent predictive carrier. On the minimal faithful spacetime branch, two logically independent finite selectors identify the same tetrahedral relation cell K_4 : exact endpoint–cycle source–rank balance among connected simple regular relations, and a stronger alternating/minimal-source selector within the pair-homogeneous simplex category. Both yield the spatial standard module

$$W_4 = \left\{ x \in \mathbb{R}^4 : \sum_{i=1}^4 x_i = 0 \right\} \cong \mathbb{R}^3.$$

A calibrated A_3 root race then reconstructs the ADM metric, lapse, and shift.

The sign and Clifford layers occur strictly later. The positive predictive word Grams and the root-race bracket cannot create a negative direction. A protected binary orientation supplies one independent signed line, and the minimal nondegenerate extension of the positive three-dimensional spatial form therefore has Lorentzian inertia $(1, 3)$, up to overall sign, *before* a Clifford representation is chosen. The represented orientation involution is

$$J = P_+ - P_-.$$

A minimal complex Clifford realization may then be chosen with generators γ^μ satisfying

$$(\gamma^0)^2 = -I, \quad (\gamma^a)^2 = I, \quad \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 0 \quad (\mu \neq \nu),$$

and hence

$$\boxed{\mathcal{A}_{\text{ext},X} := C^*(\gamma^0, \gamma^1, \gamma^2, \gamma^3) \cong M_4(\mathbb{C}).} \quad (5)$$

The corresponding principal symbol squares to the already reconstructed Lorentzian ADM quadratic form. Thus the matrix factor entering the duality is a faithful representation of a signed geometry selected upstream; it is not an algebra introduced to manufacture the spacetime signature.

For the finite theorems of Sections 2–4, the only required spacetime input is the faithfully represented Clifford factor in (5), together with the explicitly stated tetrahedral source data used later in the structural branch. Predictive reconstruction, the two K_4 selectors, root-race inversion, ADM calibration, the one-sign Lorentzian extension, and the Einstein handoff are therefore realization results rather than hidden premises of the finite duality.

2.2 Abstract spectral input and its Renewal realization

The spectral companion paper [2] gives an independent Renewal-Geometric realization of the same finite-carrier philosophy. Starting from the predictive Grand Tensor it reconstructs a finite history algebra and, from a compatible first-order variation, a real even finite spectral triple. None of that reconstruction is needed to prove the finite commutant theorems below: Definition 2.1 states the represented algebraic data axiomatically.

The spectral construction becomes relevant only for the cofinal stability problem, where it provides one mechanism producing compact Hodge/current screens. Theorem 4.46 is nevertheless stated purely in terms of Mosco convergence, compact resolvent control, and protected kernel projections; it is independent of how such screens are produced.

The joint problem is not solved by taking the direct sum of the two previously constructed sectors. A direct sum would erase precisely the mixed words that record matter incidence, gauge transport, grading change, and common-source orientation. The new object is therefore a common represented packet.

2.3 Source-complete joint packets

Definition 2.1 (Finite source-complete joint packet). At cutoff X , a finite source-complete spacetime–gauge packet consists of:

- (J1) a finite Hilbert carrier $\mathcal{H}_X$ on which the reconstructed external Clifford algebra $\mathcal{A}_{\text{ext},X} \cong M_4(\mathbb{C})$ acts faithfully;
- (J2) a commuting finite-dimensional typed internal C^* -algebra $\mathcal{A}_{\text{int},X}$;
- (J3) a central type decomposition

$$\mathcal{H}_X \cong \bigoplus_{\lambda \in \Lambda_X} \mathbb{C}^4 \otimes V_{\lambda,X} \otimes M_{\lambda,X},$$

such that

$$\mathcal{A}_{\text{int},X} = \bigoplus_{\lambda} I_4 \otimes \mathcal{B}(V_{\lambda,X}) \otimes I_{M_{\lambda,X}};$$

- (J4) a finite directed incidence graph $\Gamma_X = (\Lambda_X, E_X)$ and represented incidence histories

$$Y_{e,X} : \mathbb{C}^4 \otimes V_{s(e),X} \otimes M_{s(e),X} \longrightarrow \mathbb{C}^4 \otimes V_{t(e),X} \otimes M_{t(e),X};$$

- (J5) the mixed history words needed to determine products and adjoints of $\mathcal{A}_{\text{ext},X}$, $\mathcal{A}_{\text{int},X}$, and the $Y_{e,X}$ on one common carrier.

All marks used in a claimed mixed relation are required to occur on one future-minimal readable packet. No inverse incidence map is part of the definition.

Definition 2.2 (Joint typed action algebra). For a joint packet define

$$\boxed{\mathcal{A}_{\text{act},X} := C^*(\mathcal{A}_{\text{ext},X}, \mathcal{A}_{\text{int},X}, Y_{e,X}, Y_{e,X}^* : e \in E_X)}. \quad (6)$$

The notation “action” refers here to represented action on the common finite carrier; the variational action functional is introduced later.

Remark 2.3 (Why the joint packet is not a direct sum). The mixed incidence words are not decorative. They are exactly the data that reduce the large commutant of the bare external factor to the physically surviving multiplicity algebra. Replacing the common carrier by a direct sum of a spacetime packet and an internal packet would make (6) block diagonal and would destroy the duality proved below.

2.4 Source identifiability on the acted-on carrier

The source-complete packet is a represented system object. It cannot in general be reconstructed from a classical outcome record or from the algebra of an auxiliary environment alone. The following finite statement separates what such coarse data determine from what is needed for the duality.

Countertheorem 2.4 (Classical record data do not determine the system algebra). *There exist two finite normalized instruments on the same input and output Hilbert spaces with the same resolved classical record channel at every word length, the same one-step effects, the same one-letter Hilbert–Schmidt Gram, and the same nonzero coarse Choi spectrum, but with nonisomorphic represented system coefficient algebras and commutants. Consequently no theorem using only those record/environment invariants can identify the joint action algebra of Definition 2.2.*

Proof. An explicit pair is constructed in Appendix J. One coefficient family is simultaneously diagonal and generates $\mathbb{C}^{20}$; the other generates $M_4(\mathbb{C}) \otimes I_5$. Each resolved branch is a common scalar multiple of a unitary, so every uninterrupted classical word has the same effect in the two realizations, while the coefficient algebras and their commutants are different. The displayed Gram and Choi identities in Appendix J complete the finite witness. $\square$

Let $V : \mathcal{H} \rightarrow \mathcal{H} \otimes \mathcal{E}$ be an isometry and choose an orthonormal environment basis $(|\nu\rangle)$. Write

$$V = \sum_{\nu} C_{\nu} \otimes |\nu\rangle, \quad C(z) = (I \otimes \langle z|)V,$$

and let $\mathcal{P}_*$ be the complementary channel. Its Heisenberg pullback obeys

$$\mathcal{P}^*(|z\rangle\langle w|) = C(z)^*C(w). \quad (7)$$

Theorem 2.5 (Identity-pivot reconstruction of the system action). *Suppose the occurring coefficient algebra satisfies $\mathcal{A}_{\text{sys}} = C^*(C_{\nu} : \nu) = C^*(A_1, \dots, A_s)$ and there are environment vectors $z_0, z_1, \dots, z_s$ with*

$$C(z_0) = \alpha I, \quad C(z_j) = \beta_j A_j, \quad \alpha\beta_j \neq 0.$$

Then

$$\boxed{C^*(\mathcal{P}^*(\mathcal{B}(\mathcal{E}))) = \mathcal{A}_{\text{sys}}, \quad [\mathcal{P}^*(\mathcal{B}(\mathcal{E}))]' = \mathcal{A}'_{\text{sys}}.} \quad (8)$$

In particular, a complete operator-valued complementary pullback recovers the represented system action whenever the physical source contains such an identity pivot; no fitted remainder algebra is required.

Proof. Every pullback $C(z)^*C(w)$ belongs to $C^*(C_{\nu})$ and hence to $\mathcal{A}_{\text{sys}}$. Conversely,

$$\mathcal{P}^*(|z_0\rangle\langle z_j|) = \bar{\alpha}\beta_j A_j,$$

so every generator A_j lies in the pullback algebra. Equality of the commutants follows immediately. $\square$

Corollary 2.6 (Finite coherent system export). *For the normalized six-edge source class of Appendix J, ten fixed Hermitian environment quadratures suffice to reconstruct all six edge coefficients on the acted-on carrier. With Hilbert–Schmidt coefficient error e_K as defined in (137), the associated star-closed commutator map changes in operator norm by at most $2\sqrt{2}e_K$. Hence an independently verified protected commutant complement with measured least singular value $\hat{s} > 2\sqrt{2}e_K$ has no additional commutant direction.*

Remark 2.7 (Logical role of source identification). Theorem 2.4 is a stopping theorem, not a weakening of the double-centralizer result. Once the represented system operators are supplied or reconstructed, Theorem 3.11 is exact. What cannot be inferred from coarse record data is which system operators occurred on the common carrier in the first place.

2.5 Historical operation recognition and word-native source compilation

The preceding reconstruction problem has a second layer. Even after a finite history algebra has been reconstructed, an invariant endomorphism of a word Hilbert space need not be an operation induced by the historical system algebra. The following module criterion separates these notions.

Let $\mathcal{A} \subseteq \mathcal{B}(\mathcal{H})$ be a finite unital C^* -algebra generated by the represented historical operations on one exposed carrier, and let a finite group G act by unitaries u_g with $u_g \mathcal{A} u_g^* = \mathcal{A}$. Write $\alpha_g(a) = u_g a u_g^*$ and

$$\mathcal{A} \cong \bigoplus_{\beta} M_{n_{\beta}}(\mathbb{C}), \quad D = \dim_{\mathbb{C}} \mathcal{A} = \sum_{\beta} n_{\beta}^2,$$

with normalized regular trace

$$\tau_{\text{reg}}(a) = D^{-1} \sum_{\beta} n_{\beta} \text{Tr}(a_{\beta}).$$

On $\mathcal{W} = L^2(\mathcal{A}, \tau_{\text{reg}})$ let $L_a x = ax$, $R_b x = xb$ and $U_g x = \alpha_g(x)$.

Theorem 2.8 (Word-module recognition of historical internal actions). *The operations on the regular word carrier induced by invariant historical system elements are exactly*

$$\boxed{R(\mathcal{A})' \cap U(G)' = L(\mathcal{A}^G), \quad \mathcal{A}^G = \mathcal{A} \cap \{u_g : g \in G\}'}. \quad (9)$$

Equivalently, $T \in \text{End}(\mathcal{W})$ is left multiplication by a unique invariant historical element if and only if it commutes with every right multiplication and every U_g ; the coefficient is $T1$.

Proof. Every L_a commutes with every R_b . Conversely, if $[T, R_b] = 0$ for all b , then $T(b) = T(R_b 1) = R_b T1 = (T1)b$, hence $T = L_{T1}$. Moreover $U_g L_a U_g^{-1} = L_{\alpha_g(a)}$, so faithfulness of left multiplication gives $[U_g, L_a] = 0$ exactly when $a \in \mathcal{A}^G$. $\square$

Proposition 2.9 (Exact module and symmetry defects). *Let $(e_{\nu})_{\nu=1}^D$ be a τ_{reg} -orthonormal basis of $\mathcal{A}$ and put $a = T1$. Define*

$$\delta_R(T)^2 = \sum_{\nu=1}^D \|[T, R_{e_{\nu}}]1\|_{\tau}^2, \quad \delta_G(T)^2 = \frac{1}{|G|} \sum_g \|[U_g, T]1\|_{\tau}^2.$$

If $E_G(a) = |G|^{-1} \sum_g \alpha_g(a)$, then

$$\|T - L_a\|_{\text{HS}(\mathcal{W})}^2 = \delta_R(T)^2, \quad (10)$$

$$\|T - L_{E_G(a)}\|_{\text{HS}(\mathcal{W})} \leq \delta_R(T) + \sqrt{D/2} \delta_G(T). \quad (11)$$

Thus both defects vanish exactly on the algebra in (9).

Proof. For each basis element, $(T - L_a)e_{\nu} = T(R_{e_{\nu}}1) - R_{e_{\nu}}T1 = [T, R_{e_{\nu}}]1$; summation gives (10). Since $U_g 1 = 1$, $[U_g, T]1 = \alpha_g(a) - a$. Orthogonal group averaging gives $|G|^{-1} \sum_g \|\alpha_g(a) - a\|_{\tau}^2 = 2\|a - E_G(a)\|_{\tau}^2$. Finally $\|L_b\|_{\text{HS}(\mathcal{W})}^2 = D\|b\|_{\tau}^2$, and the triangle inequality proves (11). $\square$

Corollary 2.10 (Word-space multiplicity firewall). *Multiplicity of the conjugation representation on $L^2(\mathcal{A}, \tau_{\text{reg}})$ need not equal multiplicity on the exposed physical carrier and cannot by itself be interpreted as an executable internal factor. Any such inference must also pass Theorem 2.8. In particular, the multiplicity-birth analysis of Section 4.1.2 always refers to an actual reachable-observable physical carrier, not merely to the regular word representation. Appendix I contains an explicit twelve-root example with an invariant $M_7(\mathbb{C})$ word-space block but no physical internal $M_3(\mathbb{C})$ factor.*

We next identify a class of mixed source quantities that require no independent measurement once the relevant typing operations are already historical.

Definition 2.11 (Word-native linear shadow). A linear map $\mathcal{L} : \mathcal{A} \rightarrow \mathcal{A}$ is *word native* if

$$\boxed{\mathcal{L}(X) = \sum_{\alpha=1}^m c_{\alpha} A_{\alpha} X B_{\alpha}}, \quad (12)$$

where $A_{\alpha}, B_{\alpha} \in \mathcal{A}$ are reconstructed historical-algebra elements and the coefficients c_{α} are fixed by the declared shadow before the target conclusion is evaluated. Finite compressions, conjugations, finite group averages and finite functional-calculus operations are included whenever their defining elements belong to $\mathcal{A}$.

Theorem 2.12 (Inserted word-moment compiler). *Let $x, y, z \in \mathcal{A}$ be represented words and let $\mathcal{L}, \mathcal{M}$ be word-native maps, with $\mathcal{M}(Z) = \sum_{\beta} d_{\beta} C_{\beta} Z D_{\beta}$. Then*

$$\langle y, \mathcal{L}(x) \rangle_{\text{HS}} = \sum_{\alpha} c_{\alpha} \tau_{\text{reg}}(y^* A_{\alpha} x B_{\alpha}), \quad (13)$$

$$\langle \mathcal{L}(x), \mathcal{M}(z) \rangle_{\text{HS}} = \sum_{\alpha, \beta} \overline{c_{\alpha}} d_{\beta} \tau_{\text{reg}}(B_{\alpha}^* x^* A_{\alpha}^* C_{\beta} z D_{\beta}). \quad (14)$$

Hence every ordinary-shadow and shadow-shadow Gram block is a finite linear combination of ordinary Grand-Tensor word moments. If the historical word algebra is already flat, the reconstructed multiplication table reduces all inserted products and no deeper physical histories are required.

Proof. The formulas are the Hilbert-Schmidt inner product followed by substitution of (12). Cyclicity of τ_{reg} rewrites each summand as the inner product of two represented algebra elements. At flatness the finite word span is a $*$ -algebra, so every inserted product reduces in its known multiplication table. $\square$

Corollary 2.13 (Table-native ancestry short). *Let W_r synthesize a flat historical word bank, let $S = W_r B_S$, $T = \mathcal{L} W_r B_T$, and define*

$$K^{00} = W_r^* W_r, \quad K^{0L} = W_r^* \mathcal{L} W_r, \quad K^{LL} = W_r^* \mathcal{L}^* \mathcal{L} W_r.$$

Then

$$\boxed{G = B_S^* K^{00} B_S, \quad C = B_S^* K^{0L} B_T, \quad H = B_T^* K^{LL} B_T}, \quad (15)$$

and the ancestry short

$$\boxed{R_{T|S} = H - C^* G^{\dagger} C} \quad (16)$$

is a deterministic function of the historical word Gram, multiplication table and the fixed word-native compiler. No independently executed mixed cross-Hankel panel is required on this branch.

Countertheorem 2.14 (Word moments cannot manufacture an external shadow). *There are conservative extensions with identical historical word algebra, multiplication table, complete word-Gram hierarchy and resolved classical cylinder probabilities but different values of a proposed grading-changing or typed-incidence cross block whose defining shadow is not represented in the historical algebra. Therefore no function of the old word moments alone can recover such an external shadow.*

Proof. Tensor a faithful historical realization with an unread two-level spectator on which every old operation acts trivially. One extension contains no coherent cross-support link; another admits a future-visible spectator link and its typed compression. All old word products and moments agree, whereas the proposed external cross block is zero in the first extension and nonzero in the second. $\square$

2.6 Why the naive spacetime–gauge commutant fails

2.6.1 The factor obstruction

The simplest possible duality would identify the internal gauge algebra with the full commutant of the external Clifford factor. Finite-dimensional representation theory rules this out.

Theorem 2.15 (Factor-commutant obstruction). *Let a unital copy of $M_n(\mathbb{C})$ act on a finite-dimensional Hilbert space $\mathcal{H}$. Then for some finite multiplicity space $\mathcal{K}$,*

$$\mathcal{H} \cong \mathbb{C}^n \otimes \mathcal{K}, \quad M_n(\mathbb{C}) = M_n(\mathbb{C}) \otimes I_{\mathcal{K}}, \quad M_n(\mathbb{C})' = I_n \otimes \mathcal{B}(\mathcal{K}). \quad (17)$$

Consequently a reducible internal algebra with nontrivial centre cannot in general equal the full commutant of the external Clifford factor.

Proof. Every finite-dimensional unital representation of the simple algebra $M_n(\mathbb{C})$ is a finite direct sum of copies of its defining representation. Combining the copies gives $\mathcal{H} \cong \mathbb{C}^n \otimes \mathcal{K}$. An operator commutes with every matrix unit on the first factor if and only if it is the identity there and arbitrary on the multiplicity factor. $\square$

For the spacetime branch $n = 4$. Thus the raw Clifford commutant is always a full matrix algebra on the unresolved multiplicity space. The Standard Model cannot appear merely by writing $\mathcal{A}_{\text{SM}} = \mathcal{A}'_{\text{ext}}$. Incidence data are essential.

2.6.2 External, internal, and multiplicity factors

Theorem 2.16 (External–internal–multiplicity decomposition). *Let $\mathcal{A}_{\text{ext}} \cong M_n(\mathbb{C})$ and let $\mathcal{A}_{\text{int}}$ be a commuting finite-dimensional unital C^* -algebra on $\mathcal{H}$. Then there are finite spaces V_λ and M_λ such that*

$$\boxed{\mathcal{H} \cong \bigoplus_{\lambda \in \Lambda} \mathbb{C}^n \otimes V_\lambda \otimes M_\lambda,} \quad (18)$$

$$\boxed{\mathcal{A}_{\text{int}} = \bigoplus_{\lambda} I_n \otimes \mathcal{B}(V_\lambda) \otimes I_{M_\lambda}.} \quad (19)$$

If $\mathcal{C} = C^*(\mathcal{A}_{\text{ext}}, \mathcal{A}_{\text{int}})$, then

$$\mathcal{C} = \bigoplus_{\lambda} \mathcal{B}(\mathbb{C}^n \otimes V_{\lambda}) \otimes I_{M_{\lambda}}, \quad (20)$$

$$\mathcal{C}' = \bigoplus_{\lambda} I_{\mathbb{C}^n \otimes V_{\lambda}} \otimes \mathcal{B}(M_{\lambda}). \quad (21)$$

In particular, the residual spaces M_{λ} , rather than the raw internal representation spaces V_{λ} , carry flavour and generation multiplicity.

Proof. Apply Theorem 2.15 to write $\mathcal{H} = \mathbb{C}^n \otimes \mathcal{K}$ and $\mathcal{A}_{\text{int}} = I_n \otimes \tilde{\mathcal{A}}_{\text{int}}$. The finite-dimensional representation theorem gives

$$\mathcal{K} \cong \bigoplus_{\lambda} V_{\lambda} \otimes M_{\lambda}, \quad \tilde{\mathcal{A}}_{\text{int}} = \bigoplus_{\lambda} \mathcal{B}(V_{\lambda}) \otimes I_{M_{\lambda}}.$$

On each minimal central block the external and internal full matrix factors generate $\mathcal{B}(\mathbb{C}^n \otimes V_{\lambda})$. Schur's lemma then gives the commutant (21). $\square$

Interpretation 2.17. The theorem identifies three logically distinct structures. The external factor carries the Lorentzian Clifford action. The spaces V_{λ} carry the typed internal gauge representations. The spaces M_{λ} are what the combined external–internal action cannot see. Matter incidence can reduce this residual freedom, and the surviving reduction is precisely the dual algebra reconstructed in the next section.

3 Exact finite spacetime–gauge duality

3.1 Typed incidence and the residual multiplicity algebra

Definition 3.1 (Typed multiplicity algebra). Let $\Gamma = (\Lambda, E)$ be the incidence graph of a joint packet. Suppose the typed carrier-intertwiner space for each edge $e : \lambda \rightarrow \mu$ is one dimensional and choose a nonzero typed intertwiner D_e . Write

$$Y_e = D_e \otimes F_e, \quad F_e : M_{\lambda} \rightarrow M_{\mu}. \quad (22)$$

The *typed multiplicity algebra* is

$$\boxed{\mathcal{M}_{\text{type}} = \{(R_{\lambda})_{\lambda} : R_{\mu} F_e = F_e R_{\lambda}, \ R_{\lambda} F_e^* = F_e^* R_{\mu} \ \forall e : \lambda \rightarrow \mu\}}. \quad (23)$$

It acts on the full carrier as $\bigoplus_{\lambda} I_{\mathbb{C}^4 \otimes V_{\lambda}} \otimes R_{\lambda}$.

Lemma 3.2 (Multiplicity-one incidence factorization). *If the external–internal symmetry acts trivially on the multiplicity spaces and*

$$\dim_{\mathbb{C}} \text{Hom}_{\text{typed}}(\mathbb{C}^4 \otimes V_{\lambda}, \mathbb{C}^4 \otimes V_{\mu}) = 1,$$

then every equivariant incidence map has a unique factorization (22).

Proof. The equivariant hom-space splits as

$$\text{Hom}_{\text{typed}}((\mathbb{C}^4 \otimes V_{\lambda}) \otimes M_{\lambda}, (\mathbb{C}^4 \otimes V_{\mu}) \otimes M_{\mu}) \cong \text{Hom}_{\text{typed}}(\mathbb{C}^4 \otimes V_{\lambda}, \mathbb{C}^4 \otimes V_{\mu}) \otimes \text{Hom}(M_{\lambda}, M_{\mu}).$$

The first factor is the line $\mathbb{C} D_e$. $\square$

Theorem 3.3 (Exact edge-commutant reconstruction). *For*

$$\mathcal{A}_{\text{act}} = C^*(\mathcal{A}_{\text{ext}}, \mathcal{A}_{\text{int}}, Y_e, Y_e^* : e \in E),$$

one has

$$\boxed{\mathcal{A}'_{\text{act}} = \mathcal{M}_{\text{type}}.} \quad (24)$$

Proof. By Theorem 2.16, an operator commuting with $\mathcal{A}_{\text{ext}}$ and $\mathcal{A}_{\text{int}}$ has the form

$$X = \bigoplus_{\lambda} I_{\mathbb{C}^4 \otimes V_{\lambda}} \otimes R_{\lambda}.$$

For $Y_e = D_e \otimes F_e$,

$$XY_e - Y_eX = D_e \otimes (R_{\mu}F_e - F_eR_{\lambda}),$$

and the commutator with Y_e^* gives the adjoint intertwining equation. Because $D_e \neq 0$, the full commutators vanish exactly when (23) holds. $\square$

The theorem already gives the finite duality in one direction. The reverse direction is automatic by finite bicommutant closure, but the raw edge formula does not yet reveal how singular residues behave. We therefore give two intrinsic presentations.

3.2 Support-polar form: singular residues are allowed

Let $F : E \rightarrow H$ have polar decomposition

$$F = UP, \quad P = (F^*F)^{1/2}, \quad p = U^*U = \text{supp } P, \quad q = UU^* = \text{supp}(FF^*).$$

Lemma 3.4 (Singular polar-edge equivalence). *For $R_E \in \mathcal{B}(E)$ and $R_H \in \mathcal{B}(H)$, the equations*

$$R_H F = F R_E, \quad R_E F^* = F^* R_H$$

are equivalent to

$$\boxed{[R_E, P^2] = 0, \quad R_H U = U R_E, \quad R_E U^* = U^* R_H.} \quad (25)$$

No invertibility assumption is required.

Proof. The two incidence equations imply $F^* R_H F = F^* F R_E = R_E F^* F$, hence $[R_E, P^2] = 0$. Functional calculus gives commutation with P and its support. Multiplication by the Moore–Penrose inverse of P on its support gives the partial-isometry transport equations. The converse follows by multiplying the transport equations by P and using $[R_E, P] = 0$. $\square$

On the total multiplicity carrier $\mathcal{M} = \bigoplus_{\lambda} M_{\lambda}$, let Z_{λ} denote the vertex projections and write $F_e = U_e P_e$.

Definition 3.5 (Support-polar action algebra). Define

$$\boxed{\mathcal{O}_F^{\text{sup}} = C^*(Z_{\lambda}, F_e^* F_e, U_e, U_e^* : \lambda, e) \subseteq \mathcal{B}(\mathcal{M}).} \quad (26)$$

Theorem 3.6 (Exact support-polar duality). *For arbitrary rectangular, singular, rank-deficient, or zero incidence maps,*

$$\boxed{\mathcal{M}_{\text{type}} = (\mathcal{O}_F^{\text{sup}})', \quad \mathcal{O}_F^{\text{sup}} = \mathcal{M}'_{\text{type}}.} \quad (27)$$

Moreover,

$$\boxed{C^*(Z_\lambda, F_e, F_e^* : \lambda, e) = \mathcal{O}_F^{\text{sup}}.} \quad (28)$$

Thus a zero singular value remains a genuine support boundary and is never replaced by an inverse residue.

Proof. Commutation with the vertex projections makes an operator block diagonal. Lemma 3.4 shows that commutation with $F_e^* F_e, U_e, U_e^*$ is exactly the pair of incidence intertwining equations in Definition 3.1. This gives the first equality, and finite bicommutant closure gives the second.

For (28), $F_e^* F_e$ belongs to the edge algebra. Since its spectrum is finite, interpolate the function $g(0) = 0, g(t) = t^{-1/2}$ for $t > 0$ on that spectrum by a polynomial. Then

$$U_e = F_e g(F_e^* F_e)$$

belongs to the incidence algebra. Conversely $F_e = U_e (F_e^* F_e)^{1/2}$. $\square$

3.3 Noncommuting supports and the multiplicity quiver

Support projections themselves can fail to commute. The correct residual description is then a finite operator-space quiver, in the standard representation-theoretic sense of a quiver with linear arrow spaces [41].

Definition 3.7 (Multiplicity quiver). Let

$$\mathcal{S}_F = C^*(Z_\lambda, \text{supp}(F_e^* F_e), \text{supp}(F_e F_e^*) : \lambda, e)$$

and write its canonical finite isotypic decomposition as

$$\mathcal{M} \cong \bigoplus_{a \in \mathfrak{A}} V_a \otimes N_a, \quad \mathcal{S}_F = \bigoplus_a \mathcal{B}(V_a) \otimes I_{N_a}.$$

Let $\mathbf{T}$ be any finite $*$ -closed family of further reconstructed support metrics, polar arrows, cycle products, or action generators. For a block $T_{ba} : V_a \otimes N_a \rightarrow V_b \otimes N_b$, define

$$\mathfrak{B}_{ba}(\mathbf{T}) = \{(\varphi \otimes \text{id})(T_{ba}) : \varphi \in \mathcal{B}(V_a, V_b)^*\} \subseteq \mathcal{B}(N_a, N_b).$$

The multiplicity quiver $\mathfrak{Q}_F(\mathbf{T})$ has vertex spaces N_a and arrow spaces

$$\mathfrak{B}_{ba} = \text{span}_{T \in \mathbf{T}} \mathfrak{B}_{ba}(T).$$

Its endomorphism algebra is

$$\text{End } \mathfrak{Q}_F = \{(X_a)_a : X_b B = B X_a \text{ for every } B \in \mathfrak{B}_{ba}\}.$$

Theorem 3.8 (Noncommuting-support quiver theorem). *With the preceding notation,*

$$\boxed{C^*(\mathcal{S}_F, \mathbf{T})' \cong \text{End } \mathfrak{Q}_F(\mathbf{T}).} \quad (29)$$

If

$$(T_j)_{ba} = \sum_{\alpha} E_{\alpha}^{ba} \otimes B_{j,\alpha}^{ba}$$

in Hilbert–Schmidt orthonormal bases of $\mathcal{B}(V_a, V_b)$, then the ambient commutator form restricts to

$$q_T[(X_a)] = \sum_{j,b,a,\alpha} \|B_{j,\alpha}^{ba} X_a - X_b B_{j,\alpha}^{ba}\|_{\text{HS}}^2, \quad (30)$$

whose kernel is exactly $\text{End } \mathfrak{Q}_F$.

Proof. An operator commuting with $\mathcal{S}_F$ has the form $\bigoplus_a I_{V_a} \otimes X_a$. Expanding its commutator with every T_j in the Hilbert–Schmidt basis gives precisely the quiver intertwining relations. Orthogonality gives (30). $\square$

3.4 The connected invertible branch: polar metric and holonomy

When every residue map is invertible, the duality has a particularly geometric form.

Theorem 3.9 (Polar metric–holonomy commutant). *Assume Γ is connected, all multiplicity spaces have common dimension g , and every edge map is invertible:*

$$F_e = U_e P_e, \quad U_e \in \text{U}(g), \quad P_e \succ 0.$$

Choose a root o and a spanning tree. Let $Q_{\lambda} : M_o \rightarrow M_{\lambda}$ be the ordered product of the U_e along the tree path. For $e : \lambda \rightarrow \mu$, define on the root fibre

$$K_e = Q_{\lambda}^* P_e^2 Q_{\lambda}, \quad W_e = Q_{\mu}^* U_e Q_{\lambda},$$

and put

$$\mathcal{O}_F = C^*(K_e, W_e : e \in E) \subseteq M_g(\mathbb{C}). \quad (31)$$

Then

$$\mathcal{M}_{\text{type}}^{(o)} = \mathcal{O}'_F, \quad \mathcal{O}_F = (\mathcal{M}_{\text{type}}^{(o)})'. \quad (32)$$

The multiplicity family associated with $R \in \mathcal{O}'_F$ is

$$R_{\lambda} = Q_{\lambda} R Q_{\lambda}^*.$$

Proof. The polar-edge equations give $[R_{\lambda}, P_e^2] = 0$ and $R_{\mu} = U_e R_{\lambda} U_e^*$. Along the spanning tree these force $R_{\lambda} = Q_{\lambda} R Q_{\lambda}^*$ from the root value $R = R_o$. The metric condition becomes $[R, K_e] = 0$, while a non-tree edge gives the cycle-consistency condition $[R, W_e] = 0$. Thus the residual family is equivalent to one root operator commuting with the algebra (31). Finite bicommutant closure gives the reverse equality. $\square$

Corollary 3.10 (Flat and irreducible extremes). *Under Theorem 3.9:*

- (i) $\mathcal{M}_{\text{type}}^{(o)} = M_g(\mathbb{C})$ if and only if every K_e and W_e is scalar;
- (ii) $\mathcal{M}_{\text{type}}^{(o)} = \mathbb{C}I$ if and only if $\mathcal{O}_F = M_g(\mathbb{C})$;
- (iii) if every F_e is unitary, then $K_e = I$ and the obstruction reduces to relative cycle holonomy;
- (iv) even on a tree, a non-scalar positive polar metric can reduce the residual multiplicity algebra.

3.5 The main finite duality theorem

Theorem 3.11 (Joint action–multiplicity double-centralizer theorem). *For every finite source-complete joint packet of Definition 2.1,*

$$\boxed{\mathcal{A}'_{\text{act},X} = \mathcal{M}_{\text{type},X}, \quad \mathcal{M}'_{\text{type},X} = \mathcal{A}_{\text{act},X}.} \quad (33)$$

On the total multiplicity carrier the same dual pair has the support-exact presentation

$$\mathcal{M}_{\text{type},X} = (\mathcal{O}_{F,X}^{\text{sup}})', \quad \mathcal{O}_{F,X}^{\text{sup}} = \mathcal{M}'_{\text{type},X}.$$

If the support projections do not commute, the residual algebra is canonically represented as

$$\mathcal{M}_{\text{type},X} \cong \text{End } \mathfrak{Q}_{F,X}.$$

On a connected invertible branch, choosing one root multiplicity fibre gives

$$\mathcal{M}_{\text{type},X}^{(o)} = \mathcal{O}'_{F,X}, \quad \mathcal{O}_{F,X} = (\mathcal{M}_{\text{type},X}^{(o)})'.$$

Hence the exact duality is

$$\boxed{\text{joint typed spacetime–gauge–incidence action} \quad \longleftrightarrow \quad \text{residual gauge–matter multiplicity}.} \quad (34)$$

Proof. The first commutant equality is Theorem 3.3; the second is finite bicommutant closure. The remaining presentations are Theorems 3.6, 3.8, and 3.9. $\square$

Corollary 3.12 (Reciprocal Wedderburn structure and common centre). *For every finite joint packet satisfying Theorem 3.11, there are a finite index set $\mathfrak{I}_X$ and finite Hilbert spaces $H_{\alpha,X}^{\text{act}}$ and $H_{\alpha,X}^{\text{mult}}$ such that, after one unitary decomposition of the represented carrier,*

$$\boxed{\mathcal{H}_X \cong \bigoplus_{\alpha \in \mathfrak{I}_X} H_{\alpha,X}^{\text{act}} \otimes H_{\alpha,X}^{\text{mult}},} \quad (35)$$

with

$$\mathcal{A}_{\text{act},X} \cong \bigoplus_{\alpha \in \mathfrak{I}_X} \mathcal{B}(H_{\alpha,X}^{\text{act}}) \otimes I_{H_{\alpha,X}^{\text{mult}}}, \quad (36)$$

$$\mathcal{M}_{\text{type},X} \cong \bigoplus_{\alpha \in \mathfrak{I}_X} I_{H_{\alpha,X}^{\text{act}}} \otimes \mathcal{B}(H_{\alpha,X}^{\text{mult}}). \quad (37)$$

In particular,

$$\boxed{Z(\mathcal{A}_{\text{act},X}) = Z(\mathcal{M}_{\text{type},X}) = \mathcal{A}_{\text{act},X} \cap \mathcal{M}_{\text{type},X}.} \quad (38)$$

The common minimal central projections therefore label the same represented sectors on both sides, while the two matrix sizes are reciprocal action and multiplicity dimensions inside each sector. Any unitary equivalence of source-complete joint packets transports the two algebras and their common centre simultaneously.

Proof. Apply the finite-dimensional structure theorem to the unital C^* -algebra $\mathcal{A}_{\text{act},X}$. In a unitary decomposition of its representation it has the form $\bigoplus_{\alpha} \mathcal{B}(H_{\alpha,X}^{\text{act}}) \otimes I$. Its commutant is therefore exactly the algebra in (37), which equals $\mathcal{M}_{\text{type},X}$ by Theorem 3.11. The centre of either algebra is generated by the same minimal central block projections, proving (38). Simultaneous unitary covariance is immediate from the defining commutant identities. $\square$

3.6 Incidence essentiality and source-minimal skeletons

Fix the non-incidence action $\mathcal{A}_0 = C^*(\mathcal{A}_{\text{ext}}, \mathcal{A}_{\text{int}})$ and a finite incidence bank E . For $S \subseteq E$ set

$$\mathcal{M}(S) = C^*(\mathcal{A}_0, Y_e, Y_e^* : e \in S)', \quad \mathcal{M} = \mathcal{M}(E),$$

and define the residual test space $V = \mathcal{A}'_0 \cap \mathcal{M}^\perp$, with $n = \dim_{\mathbb{C}} V$. For $X \in V$ let

$$D_e X = 2^{-1/2}([Y_e, X], [Y_e^*, X]), \quad G_e = D_e^* D_e, \quad G_S = \sum_{e \in S} G_e. \quad (39)$$

The full bank is exact precisely when $G_E \succ 0$ on V .

Theorem 3.13 (Incidence rank polymatroid). *Let $R_e = \text{Ran } D_e^* \subseteq V$ and*

$$f(S) = \dim_{\mathbb{C}} \left(\sum_{e \in S} R_e \right). \quad (40)$$

Then

$$\boxed{f(S) = \text{rank } G_S = n - \dim_{\mathbb{C}}(\mathcal{M}(S) \cap V), \quad \mathcal{M}(S) = \mathcal{M} \oplus \text{Ker } G_S} \quad (41)$$

as orthogonal vector spaces. The function f is normalized, monotone and submodular:

$$f(S) + f(T) \geq f(S \cup T) + f(S \cap T). \quad (42)$$

An incidence $e \in E$ is essential for the full duality exactly when

$$\boxed{f(E \setminus \{e\}) < n} \quad (43)$$

or, equivalently, when deleting e strictly enlarges the commutant.

Proof. The quadratic form of G_S is $\sum_{e \in S} \|D_e X\|^2$, hence

$$\text{Ker } G_S = \bigcap_{e \in S} \text{Ker } D_e = \left(\sum_{e \in S} \text{Ran } D_e^* \right)^\perp.$$

Every element of $\mathcal{M}(S)$ already lies in $\mathcal{A}'_0$. Orthogonal projection onto $\mathcal{M}$ therefore leaves a residual vector in V , and the incidence constraints on that residual are exactly $G_S X = 0$. This proves (41). If $R(S) = \sum_{e \in S} R_e$, then $R(S) + R(T) = R(S \cup T)$ and $R(S \cap T) \subseteq R(S) \cap R(T)$. The dimension identity for two subspaces yields submodularity. The deletion criterion is the kernel formula with $S = E \setminus \{e\}$. $\square$

Corollary 3.14 (Finite source-minimal incidence skeleton). *Every finite exact incidence bank contains an inclusion-minimal subset $S \subseteq E$ with $\mathcal{M}(S) = \mathcal{M}(E)$. Such a subset is obtained by repeatedly deleting nonessential incidences and has at most n members. In general these minimal subsets need not be unique and need not have equal cardinality; the correct grouped rank object is the polymatroid f , not a matroid on the incidence labels.*

Proof. Finite reverse deletion terminates. In an inclusion-minimal sufficient bank, adding its members one by one strictly increases the span of the corresponding constraint ranges, so at most n additions are possible. The remaining claims follow from the grouped nature of the subspaces R_e ; an explicit unequal-cardinality example is given in Appendix H. $\square$

Interpretation 3.15 (Meaning of spacetime–gauge Howe-type duality). The phrase “Howe-type duality” is used below for the mutual-centralizer content of Theorem 3.11 together with the reciprocal block structure of Corollary 3.12. The statement does not exchange a bare spacetime algebra with a bare gauge algebra. It says that, once spacetime, typed gauge action, and physical matter incidence are retained on one represented carrier, the complete visible action and the invisible matter multiplicity determine one another exactly.

Remark 3.16 (Relation to earlier commutant dualities). Classical Howe duality concerns mutually centralizing actions with strong multiplicity correspondence properties [43]. Gauge/observable Howe pairs also occur in four-dimensional conformal field theory [28]. In noncommutative Standard-Model geometry, the Hodge or Morita condition instead compares the spectral Clifford algebra with its opposite/commutant [17, 18]. Theorem 3.11 is a different finite operator-algebraic analogue: its content is the exact double-centralizer relation between the *joint* spacetime–gauge–incidence action and the residual typed multiplicity algebra on the source-complete predictive carrier. We do not assume that the pair arises from a classical reductive dual pair.

4 Structural Standard Model, coercive stability, and common-action compatibility

4.1 The minimal structural Standard-Model branch

We now identify one certified finite branch on which the residual multiplicity algebra and its incidence data have the structural Standard-Model content used below, together with a rank-three generation factor. The construction is intentionally separated from the duality theorem: the latter holds for every source-complete typed packet, whereas the present section classifies one particular finite active residual by explicit occurrence and positivity conditions.

4.1.1 Geometry shorting and the twelve-dimensional residual

Let Φ be the twelve oriented A_3 roots associated with the tetrahedral cell and let

$$P = \mathbb{R}^\Phi, \quad P_0 = \left\{ f \in P : \sum_{\alpha \in \Phi} f(\alpha) = 0 \right\}.$$

Let $\mathcal{T} = \{c, \ell\}$ be two accepted categorical types and set

$$t_0 = 2^{-1/2}(\mathbf{e}_c + \mathbf{e}_\ell), \quad t_1 = 2^{-1/2}(\mathbf{e}_c - \mathbf{e}_\ell).$$

After removal of the overall accepted mean, define

$$\boxed{E_{\text{acc}}^0 = (P_0 \otimes t_0) \oplus (\mathbb{R}\mathbf{1}_\Phi \otimes t_1) \oplus (P_0 \otimes t_1).} \quad (44)$$

The three summands have dimensions 11, 1, 11.

Write

$$E_{\text{ST}}^{\text{cat}} := P_0 \otimes t_0, \quad E_{\text{int}}^{\text{cat}} := (\mathbb{R}\mathbf{1}_\Phi \otimes t_1) \oplus (P_0 \otimes t_1).$$

Let

$$S_{\text{ST}} : E_{\text{ST}} \rightarrow \mathcal{H}_{\text{acc}}, \quad S_{\text{int}} : E_{\text{int}}^{\text{cat}} \rightarrow \mathcal{H}_{\text{acc}}$$

denote respectively the complete represented spacetime source and the accepted internal categorical source on one common carrier. The spacetime source may contain, in addition to $E_{\text{ST}}^{\text{cat}}$, independently reconstructed waiting-time, calibration, bracket, shift, or geometric-circulation directions. Put

$$G_{\text{ST}} = S_{\text{ST}}^* S_{\text{ST}}, \quad G_{\text{int}} = S_{\text{int}}^* S_{\text{int}}, \quad C = S_{\text{int}}^* S_{\text{ST}},$$

and let P_{ST} be the orthogonal projection onto $\text{Ran } S_{\text{ST}}$.

Theorem 4.1 (Universal geometry short and active residual). *For every source-complete common realization, shorting the accepted internal source by the complete represented spacetime source gives*

$$\boxed{G_{\text{int}|\text{ST}} = G_{\text{int}} - C G_{\text{ST}}^\dagger C^* = S_{\text{int}}^* (I - P_{\text{ST}}) S_{\text{int}} \succeq 0.} \quad (45)$$

The source-minimal internal categorical carrier is therefore

$$\boxed{E_{\text{int}|\text{ST}}^{\text{min}} = E_{\text{int}}^{\text{cat}} / \text{Ker } G_{\text{int}|\text{ST}}, \quad \dim E_{\text{int}|\text{ST}}^{\text{min}} = \text{rank } G_{\text{int}|\text{ST}}.} \quad (46)$$

On the explicit orthogonal active realization, the accepted contrast synthesis has scalar Gram

$$S_{\text{acc}}^* S_{\text{acc}} = \vartheta I_{E_{\text{acc}}^0}, \quad \vartheta > 0,$$

and the additional spacetime scores are orthogonal to the accepted categorical source. Hence the mixed block vanishes on $E_{\text{int}}^{\text{cat}}$ and

$$\boxed{G_{\text{int}|\text{ST}} = \vartheta I_{12}.} \quad (47)$$

On the tetrahedrally equivariant branch, the categorical residual carries the multiplicity-free representation

$$\boxed{E_{\text{int}}^{\text{cat}} \otimes \mathbb{C} \cong \mathbb{C}_{\text{type}}^+ \oplus W_+ \oplus (V_2)_+ \oplus W_- \oplus (W \otimes \text{sgn})_-} \quad (48)$$

Here $\dim W = 3$ and $\dim V_2 = 2$. The odd standard block W_- is the internal endpoint module, while the odd sign-twisted block is the distinct loop module. If the restriction of $G_{\text{int}|\text{ST}}$ to the endpoint block is strictly positive, that block survives the source-minimal quotient at complex rank three.

Proof. The first identity is the Schur/Anderson–Trapp short of the joint source Gram; equivalently, it is the Gram of the orthogonal residual $(I - P_{\text{ST}})S_{\text{int}}$. Positivity and the quotient/rank statement follow immediately. On the explicit active product carrier the accepted categorical summands are orthogonal with common weight ϑ , and all additional spacetime scores lie in the independently adjoined relational port, giving (47).

The representation decomposition is the $S_4 \times \langle \iota \rangle$ decomposition of the twelve oriented roots: the reversal-even six-dimensional sector is $\mathbf{1} \oplus W \oplus V_2$, while the reversal-odd sector is $W \oplus (W \otimes \text{sgn})$. Removing the even root mean and adjoining the scalar type contrast gives (48). On an equivariant common carrier the spacetime range, its orthogonal projection, and hence the shorted Gram are equivariant, so the inequivalent endpoint and loop irreducibles remain reducing. Strict positivity on the endpoint block prevents its collapse in (46). $\square$

The essential point is that the rank-three internal endpoint block is defined *after* the complete geometric source has been removed on the same joint Gram. It is therefore neither a second name for the spatial W_4 carrier nor a dimension count copied from the external tetrahedral representation. The scalar active realization provides a particularly transparent nonempty branch, but the source-short formulation itself is not tied to an isotropic accepted metric.

4.1.2 External multiplicity census and the first internal-factor obstruction

The external tetrahedral action and the internal gauge action must be kept on opposite sides of the centralizer. Let $\mathcal{V}_r$ be the future-minimal reachable–observable residual *physical carrier* spanned by represented source words of length at most r , after the complete spacetime and physical-null shorts. This is not the regular word Hilbert space $L^2(\mathcal{A}, \tau_{\text{reg}})$: by Corollary 2.10, large invariant multiplicities of the latter do not by themselves create physical internal factors. Decompose the external S_4 action as

$$\mathcal{V}_r \cong \bigoplus_{\lambda \in \widehat{S}_4} V_\lambda \otimes M_{\lambda,r}, \quad m_{\lambda,r} := \dim_{\mathbb{C}} M_{\lambda,r}. \quad (49)$$

Theorem 4.2 (External isotypic algebra and multiplicity census). *At every finite word depth,*

$$\mathcal{A}_{\text{ext},r} = C^*(\rho_r(S_4)) \cong \bigoplus_{\lambda} M_{\dim V_\lambda}(\mathbb{C}) \otimes I_{M_{\lambda,r}}, \quad (50)$$

$$\mathcal{A}'_{\text{ext},r} \cong \bigoplus_{\lambda} I_{V_\lambda} \otimes M_{m_{\lambda,r}}(\mathbb{C}). \quad (51)$$

Consequently an internal simple factor $M_n(\mathbb{C})$ can occur on the relative commutant side only if some external irreducible occurs with multiplicity at least n .

On the categorical degree-one carrier,

$$\mathcal{V}_{\text{cat}} \cong \mathbb{C}_{\text{type}} \oplus (W \otimes \mathbb{C}^2) \oplus V_2 \oplus W^{\text{sgn}},$$

so the multiplicities are $(1, 2, 1, 1)$ and

$$\boxed{\mathcal{A}'_{\text{ext},1} \cong \mathbb{C} \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}.} \quad (52)$$

If one additional external-trivial private contrast line is retained, the multiplicity vector becomes $(2, 2, 1, 1)$ and the relative commutant is

$$M_2(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}. \quad (53)$$

Neither algebra contains a unital copy of $M_3(\mathbb{C})$.

Proof. Finite-dimensional isotypic decomposition gives (49). Burnside’s theorem gives the full matrix algebra on each irreducible external factor and the identity on its multiplicity factor; Schur’s lemma gives the commutant. A unital $*$ -representation of $M_n(\mathbb{C})$ is a direct sum of n -dimensional defining representations, so no simple commutant block of size smaller than n can contain it. The two displayed finite censuses follow from the stated multiplicities. $\square$

Countertheorem 4.3 (Degree-one internal-colour obstruction). *The populated categorical degree-one residual cannot by itself support a commuting internal colour factor $M_3(\mathbb{C})$. The same remains true after adjoining only one external-trivial private contrast line.*

Proof. This is the final assertion of Theorem 4.2: every simple summand of the corresponding relative commutant has size at most two. $\square$

Let K_r denote the labelled source-word Gram whose quotient span is $\mathcal{V}_r$, and define the external-trivial multiplicity

$$m_{0,r} := \dim_{\mathbb{C}} \text{Hom}_{S_4}(\mathbf{1}, \mathcal{V}_r). \quad (54)$$

The first colour-eligible depth is

$$r_{\text{col}}^{\text{birth}} := \min\{r : m_{0,r} \geq 3\}, \quad (55)$$

with value ∞ if no such depth occurs before the word hierarchy becomes flat.

Theorem 4.4 (Flat-depth multiplicity-birth obstruction). *Suppose*

$$\text{rank } K_r = \text{rank } K_{r+1} \quad (56)$$

and $m_{0,r} \leq 2$. Then $\mathcal{V}_s = \mathcal{V}_r$ for every $s \geq r$ and no later physical source word can produce an external-trivial internal $M_3(\mathbb{C})$ factor. In particular $r_{\text{col}}^{\text{birth}} = \infty$ on that source algebra.

Proof. By construction $\mathcal{V}_{r+1}$ is the span of $\mathcal{V}_r$ and all one-letter images of $\mathcal{V}_r$. Equality of Gram ranks therefore gives $\mathcal{V}_{r+1} = \mathcal{V}_r$, so $\mathcal{V}_r$ is invariant under every source letter. Induction gives $\mathcal{V}_s = \mathcal{V}_r$ for all later s . The multiplicity census is consequently frozen, and Theorem 4.2 excludes an internal $M_3(\mathbb{C})$ block. $\square$

A positive multiplicity birth is necessary but not sufficient for colour: it creates only a coherent multiplicity carrier. The physical internal word algebra must still connect its minimal projections.

4.1.3 Source-minimal routers and exact internal landing

Theorem 4.5 (Connected-router full-matrix theorem). *Let*

$$\mathcal{D}_n = \text{span}\{p_1, \dots, p_n\} \cong \mathbb{C}^n$$

be the diagonal algebra of mutually orthogonal rank-one projections on $\mathbb{C}^n$, and let $u \in M_n(\mathbb{C})$. Form the undirected support graph with an edge $i - j$ whenever $p_i u p_j \neq 0$ or $p_j u p_i \neq 0$. Then

$$\boxed{C^*(\mathcal{D}_n, u) = M_n(\mathbb{C}) \iff \text{the support graph of } u \text{ is connected.}} \quad (57)$$

If the graph is disconnected, every connected component is a reducing subspace and the generated algebra is a proper block-diagonal subalgebra.

Proof. For every support edge, $p_i u p_j$ is a nonzero scalar multiple of the matrix unit E_{ij} ; adjoints give E_{ji} . Products along a graph path generate all matrix units between vertices in the same connected component. Thus a connected graph generates all of $M_n(\mathbb{C})$. Conversely the sum of the p_i over any connected component is a nontrivial central reducing projection when the graph is disconnected. $\square$

Corollary 4.6 (One-router source minimality). *Once a coherent n -dimensional multiplicity carrier and its diagonal minimal projections have occurred, one cyclic nonnormal router is sufficient to generate $M_n(\mathbb{C})$. Without any off-diagonal word the generated algebra remains commutative. Thus one router is source-minimal at fixed multiplicity carrier.*

Let $T \cong \mathbb{C}$ be the surviving external-trivial type line and let $H_{\text{priv}} \cong \mathbb{C}^2$ be a faithful external-trivial private plane on the same source carrier. Let p_T and p_H denote the corresponding projections. On the standard-isotypic multiplicity plane let p_t and p_h be rank-one projections onto unit vectors t and h .

Lemma 4.7 (Weak multiplicity router). *If*

$$0 < |\langle t, h \rangle| < 1, \quad (58)$$

then

$$\boxed{C^*(p_t, p_h) = M_2(\mathbb{C}).} \quad (59)$$

With only one rank-one projection the generated algebra is $\mathbb{C}^2$.

Proof. The two projections have no common one-dimensional reducing subspace, so Burnside's theorem gives $M_2(\mathbb{C})$. A single projection generates only its two spectral projections. $\square$

For an actual operation word u on $T \oplus H_{\text{priv}}$ define

$$\omega_{\text{col}}(u) = \|p_H u p_T\|_{\text{HS}}^2 + \|p_T u p_H\|_{\text{HS}}^2. \quad (60)$$

Theorem 4.8 (Necessary and sufficient colour-bridge criterion). *Assume the represented private support words generate $M_2(H_{\text{priv}})$. Then*

$$\boxed{C^*(\mathbb{C}p_T \oplus M_2(H_{\text{priv}}), u) = M_3(\mathbb{C}) \iff \omega_{\text{col}}(u) > 0.} \quad (61)$$

In the zero-bridge case the algebra is exactly $\mathbb{C} \oplus M_2(\mathbb{C})$ and has a nontrivial central projection.

Proof. A nonzero corner between the type line and the private 2×2 block, together with its adjoint and the private matrix units, generates every rectangular matrix unit and hence $M_3(\mathbb{C})$. If both cross corners vanish, the line and plane remain reducing and the algebra is precisely the direct sum $\mathbb{C} \oplus M_2(\mathbb{C})$. $\square$

The bridge need not be a primitive operation when the historical route table contains reciprocal access through one common mediator. Let G be the external finite symmetry, let V be an irreducible unitary G -module of dimension d , and let the mediator carrier be $\mathcal{E} = V \otimes \mathcal{N}$ with $n = \dim \mathcal{N}$. Suppose actual entry and reverse-exit operations have the covariant form

$$A_g = (v_g \otimes I)A : T \rightarrow \mathcal{E}, \quad B_h^* = B^*(v_h^* \otimes I) : \mathcal{E} \rightarrow H_{\text{priv}},$$

and that the actual mediator dynamics contains $D = I_V \otimes h$ with $h = h^*$. Put

$$r_{g,k} = B^*(v_g \otimes h^k)A, \quad 0 \leq k < n,$$

$$\rho_A = \text{Tr}_V(AA^*), \quad \rho_B = \text{Tr}_V(BB^*), \quad W_A = \sum_{k=0}^{n-1} h^k \rho_A h^k.$$

Theorem 4.9 (Finite reciprocal-return criterion). *For the declared route bank,*

$$\boxed{\mathbf{m}_{\text{ret}} := \frac{1}{|G|} \sum_{g \in G} \sum_{k=0}^{n-1} \|r_{g,k}\|_{\text{HS}}^2 = \frac{1}{d} \text{Tr}(\rho_B W_A) \geq 0.} \quad (62)$$

The following are equivalent:

- (i) *an admitted word in the route bank has a nonzero $H_{\text{priv}} \leftarrow T$ corner;*
- (ii) *some $r_{g,k}$ is nonzero;*

(iii) $\mathbf{m}_{\text{ret}} > 0$;

(iv) the h -cyclic multiplicity spaces generated by $\text{Ran } \rho_A$ and $\text{Ran } \rho_B$ are not orthogonal.

If these conditions fail, an explicit reducing projection separates T from H_{priv} for the entire declared bank, so arbitrary repeated excursions cannot create a colour bridge within that bank. If they hold, a witness has at most $n + 1$ route letters.

Proof. Schur averaging on the irreducible V factor gives

$$\frac{1}{|G|} \sum_g (v_g \otimes I) A A^* (v_g^* \otimes I) = \frac{I_V}{d} \otimes \rho_A.$$

Taking the Hilbert–Schmidt norm of $B^*(v_g \otimes h^k)A$ and summing yields (62). Since it is a finite sum of squared norms, positivity is equivalent to a nonzero returned word. The range of W_A is $\text{span}\{h^k \text{Ran } \rho_A : 0 \leq k < n\}$ by Cayley–Hamilton. Thus $\text{Tr}(\rho_B W_A) = 0$ exactly when the two cyclic multiplicity spaces are orthogonal. In that case the projection onto $T \oplus (V \otimes \text{Ran } W_A)$ commutes with the route generators and their adjoints while annihilating H_{priv} , excluding all longer returns. The finite horizon $k < n$ gives the stated word-length bound. $\square$

Corollary 4.10 (Port-relative reachability forces colour). *If $W_A \succeq \kappa I_{\mathcal{N}}$ with $\kappa > 0$ and $B \neq 0$, then*

$$\boxed{\mathbf{m}_{\text{ret}} \geq \frac{\kappa}{d} \|B\|_{\text{HS}}^2, \quad \max_{g, k < n} \|r_{g,k}\|_{\text{HS}}^2 \geq \frac{\kappa}{dn} \|B\|_{\text{HS}}^2.} \quad (63)$$

Hence the colour bridge follows from the upstream route structure and Theorem 4.8. In particular, for a multiplicity-one mediator, two nonzero reciprocal ports force a two-leg colour return and no independent direct bridge is required.

Blockwise matrix generation still does not by itself guarantee that all central blocks are independent. The only ambiguity after the nonisomorphic M_3 and M_2 blocks have landed concerns the two scalar sectors.

Theorem 4.11 (Central separation of the landed internal blocks). *Let $\mathcal{B} \subseteq M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}$ be a unital C^* -subalgebra whose projection onto each displayed factor is surjective. Then exactly one of the following occurs:*

$$\mathcal{B} = M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}, \quad \dim_{\mathbb{C}} \mathcal{B} = 15, \quad (64)$$

$$\mathcal{B} = \{(A, B, z, z) : A \in M_3(\mathbb{C}), B \in M_2(\mathbb{C}), z \in \mathbb{C}\}, \quad \dim_{\mathbb{C}} \mathcal{B} = 14. \quad (65)$$

If χ_1, χ_2 are the scalar characters and (b_j) is any generating bank, then the full product occurs exactly when

$$\boxed{\eta_{\text{cen}} := \sum_j |\chi_1(b_j) - \chi_2(b_j)|^2 > 0.} \quad (66)$$

In particular, represented central projectors onto the two scalar sectors discharge the condition automatically.

Proof. A unital subdirect product of finite simple matrix algebras can correlate only isomorphic simple factors; a proof is given in Appendix I. The M_3 and M_2 factors are therefore independent and only the two scalar factors may be identified. They are independent exactly when their characters differ on the generated algebra, which is equivalent to (66). $\square$

Define $\mathcal{A}_{\text{int},r}^{\text{word}}$ to be the C^* -algebra generated by all represented physical word compressions at depth r that commute with the external tetrahedral action.

Theorem 4.12 (Relative-commutant saturation). *Suppose that at one finite word depth the relevant external multiplicity vector is*

$$(m_0, m_W, m_{V_2}, m_{W^{\text{sgn}}}) = (3, 2, 1, 1), \quad (67)$$

that the external-trivial three-space is coherently realized as $T \oplus H_{\text{priv}}$, and that:

- (I1) *the private support words generate $M_2(H_{\text{priv}})$ and either one represented word has $\omega_{\text{col}} > 0$, or an admitted reciprocal mediator bank satisfies $\mathbf{m}_{\text{ret}} > 0$ in Theorem 4.9;*
- (I2) *the standard multiplicity plane contains two distinct nonorthogonal represented rank-one projections;*
- (I3) *the V_2 and W^{sgn} external blocks have multiplicity one;*
- (I4) *the two scalar sectors are centrally separated, equivalently the generated algebra contains their independent central supports or passes $\eta_{\text{cen}} > 0$ in Theorem 4.11.*

Then

$$\boxed{\mathcal{A}_{\text{int},r}^{\text{word}} = \mathcal{A}'_{\text{ext},r} \cong M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}.} \quad (68)$$

Its complex dimension is 15, its centre has dimension four, and the two nonabelian simple factors are intrinsically distinguished by their matrix sizes. Hence the labels “colour” and “weak” are canonical up to inner automorphism once the multiplicity packet has landed.

Proof. Condition (I1) and Theorem 4.8 generate the M_3 block; when the reciprocal route alternative is used, Theorem 4.9 supplies the required represented bridge. Condition (I2) and Lemma 4.7 generate the M_2 block. The multiplicity-one sectors give surjective scalar compressions, and condition (I4) excludes their diagonal identification by Theorem 4.11. Thus the generated internal algebra is the full product, which is exactly the relative commutant determined by the multiplicity census in Theorem 4.2. $\square$

Theorem 4.13 (Exact internal-algebra deficit alternatives). *On a coherent carrier with multiplicity census $(3, 2, 1, 1)$, assume the colour three-space initially carries $M_2(\mathbb{C}) \oplus \mathbb{C}$, the weak multiplicity plane carries one rank-one projection, and the displayed colour, weak and two scalar central supports occur independently in the represented internal algebra. Then exactly the following dimensions occur:*

<i>represented internal words</i>	$\dim_{\mathbb{C}} \mathcal{A}_{\text{int}}^{\text{word}}$
<i>neither colour bridge nor weak router</i>	9
<i>weak router but no colour bridge</i>	11
<i>colour bridge but no weak router</i>	13
<i>both bridge and router</i>	15

The terminal value 15 is equivalent to saturation of the complete relative commutant in Theorem 4.12. Each failure is therefore an exact finite source-word rank deficit rather than a discarded branch. If independent central supports have not yet been established, Theorem 4.11 must be applied before componentwise dimensions are added; in particular, the terminal blockwise-surjective branch may still be the 14-dimensional scalar-locked algebra (65).

Proof. Without the bridge the colour block is $M_2 \oplus \mathbb{C}$, of dimension five; with the bridge it is M_3 , of dimension nine. One weak rank-one projection generates $\mathbb{C}^2$, of dimension two; a noncommuting second projection generates M_2 , of dimension four. The two multiplicity-one sectors add two scalar dimensions, giving 9, 11, 13, 15. $\square$

Corollary 4.14 (Generic saturation at fixed multiplicity census). *Fix the multiplicity census $(3, 2, 1, 1)$ and the diagonal minimal projections on the 3- and 2-dimensional multiplicity carriers. In the affine parameter space of internal router operators on those carriers, the set that generates $M_3(\mathbb{C}) \oplus M_2(\mathbb{C})$ is Zariski open and dense. Its complement is a finite union of proper linear subspaces corresponding to disconnected support graphs. This genericity is conditional on the multiplicity carriers having occurred; it does not assert that a microscopic source samples the full router parameter space.*

Proof. For fixed n , failure of connectedness means that there exists a nontrivial partition of $\{1, \dots, n\}$ across which all matrix entries of the router vanish in both directions. For each partition this is a proper linear subspace of $M_n(\mathbb{C})$, and there are finitely many partitions. The complement is therefore Zariski open and dense. Apply this independently to $n = 3$ and $n = 2$ and use Theorem 4.5. $\square$

Remark 4.15 (Selection content of the internal landing). The multiplicity census, bridge and router conditions are not names for a preselected gauge group. Theorem 4.3 excludes colour at the populated degree-one census, Theorem 4.4 gives a terminating no-go when multiplicity birth never occurs, and Theorem 4.13 returns the exact smaller algebra whenever one of the two connecting words is absent. Conversely, once the required multiplicity carriers exist, one bridge and one router are source-minimal for the nonabelian blocks, and on reciprocal-route branches the bridge itself is forced by the positive return criterion rather than separately postulated. Full product saturation additionally requires the finite central-separation test of Theorem 4.11. The determinant test below is applied only after this multiplicity-side landing; applying it to the external Burnside blocks would act on the wrong side of the centralizer.

Remark 4.16 (Relation to the finite algebra of almost-commutative geometry). The algebra in (68) is the complex relative commutant of the present joint action. It should not be identified with the real finite coordinate algebra used in the standard almost-commutative description of the Standard Model. The two constructions organize the internal sector differently even though they recover the same physical nonabelian dimensions and quotient gauge group on the branch described below.

4.1.4 The determinant seed and the global gauge group

The global form of the Standard-Model gauge group is not fixed by its Lie algebra alone. In conventional field theory one may quotient $SU(3) \times SU(2) \times U(1)$ by different subgroups of the common $\mathbb{Z}_6$ centre, with physically distinct line-operator spectra [29, 30]. The statement below is therefore branch-specific: the determinant seed reconstructs the faithful $\mathbb{Z}_6$ quotient rather than assuming it as the definition of the Standard Model.

Let $C \cong \mathbb{C}^3$ and $W_2 \cong \mathbb{C}^2$ denote the nonabelian multiplicity carriers after the relative-commutant landing of Theorem 4.12. The determinant relation is not inserted as an independent group constraint, nor is it applied to the external tetrahedral Burnside blocks. It is the nonzero branch of a canonical alternating projection of the represented incidence tensor on the landed multiplicity carriers.

Proposition 4.17 (Canonical determinant incidence). *Consider the represented incidence space*

$$\text{Hom}(C \otimes W_2 \otimes W_2, \Lambda^2 C^*).$$

Its component that is fully alternating in the three colour and two weak arguments is one-dimensional and canonically isomorphic to

$$(\det C \otimes \det W_2)^*.$$

Consequently every represented incidence tensor has a canonical alternating projection τ . If $\tau \neq 0$, it defines a determinant seed uniquely up to a nonzero scalar and, after unit normalization, up to phase. All nonzero normalized choices have the same unitary stabilizer. If the alternating projection vanishes, the determinant-preserving branch is absent rather than being supplied by completion.

Proof. Complete alternation is the equivariant projection onto the unique $(\det C \det W_2)^{-1}$ symmetry line. The explicit incidence map and normalization are recorded in Appendix D; in particular, Proposition D.1 identifies this line isomorphically with $(\det C \otimes \det W_2)^*$. $\square$

Definition 4.18 (Determinant seed). On the nonzero branch of Proposition 4.17, normalize its alternating projection to a unitary isomorphism

$$\tau : \det C \otimes \det W_2 \xrightarrow{\sim} \mathbb{C}. \quad (69)$$

Only the line and its stabilizer are physically relevant; multiplying τ by an overall phase leaves the stabilizer unchanged. Thus τ is a reconstructed incidence shadow on this branch, not an independently chosen Standard-Model datum.

Theorem 4.19 (Gauge-group reconstruction). *The unitary automorphism group of the determinant seed is*

$$\boxed{\text{Aut}_{\text{U}}(C, W_2, \tau) = S(\text{U}(3) \times \text{U}(2)) \cong \frac{\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)}{\mathbb{Z}_6}}. \quad (70)$$

Proof. A pair $(A, B) \in \text{U}(C) \times \text{U}(W_2)$ preserves τ exactly when $\det A \det B = 1$. The map

$$\Phi(g_3, g_2, z) = (z^{-2}g_3, z^3g_2)$$

has image in this subgroup and kernel

$$\{(z^2I_3, z^{-3}I_2, z) : z^6 = 1\} \cong \mathbb{Z}_6.$$

$\square$

Proposition 4.20 (Positive operational source criterion for the determinant group). *Let $\chi(A, B) = \det A \det B$ on $\text{U}(C) \times \text{U}(W_2)$. Suppose an actual typed incidence instrument contains two marked coherent amplitude sectors on which this group acts by the characters 1 and χ^{-1} . Let J_{01} denote the corresponding Choi cross block and put*

$$\boxed{h_{\det} := \|J_{01}\|_{\text{HS}}^2}. \quad (71)$$

If $h_{\det} > 0$, the stabilizer of the marked positive subpacket inside $\text{U}(3) \times \text{U}(2)$ is exactly

$$\boxed{\ker \chi = S(\text{U}(3) \times \text{U}(2))}. \quad (72)$$

If $h_{\det} = 0$, the two diagonal character-sector masses alone do not fix their relative phase and therefore do not imply the determinant subgroup.

Proof. Under (A, B) the cross block transforms by the relative character $\chi(A, B)^{-1}$. A nonzero marked cross block is therefore fixed exactly when $\chi(A, B) = 1$. If the cross block vanishes, independent phase rotation of the two diagonal sectors leaves their positive masses unchanged. $\square$

Remark 4.21 (Determinant incidence is genuinely higher arity). The determinant character cannot be manufactured by arbitrary temporal depth inside a single native carrier containing only the defining colour and weak blocks and neutral sectors. Appendix I proves a sharp charged-slot bound: the determinant mode is absent below five total defining tensor slots and occurs at five. The incidence space in Proposition 4.17 has precisely three colour and two weak charged legs, so its arity is minimal for this character. This is an occurrence statement, not an assertion that every microscopic source supplies the required five-leg coherent port.

Proposition 4.22 (Faithful matter realization of the determinant quotient). *Normalize the abelian charge by $y = 6Y$ and write all fermions as left-handed Weyl fields. On one generation the represented matter packet is*

$$\begin{aligned} Q : (\mathbf{3}, \mathbf{2})_1, \quad u^c : (\bar{\mathbf{3}}, \mathbf{1})_{-4}, \quad d^c : (\bar{\mathbf{3}}, \mathbf{1})_2, \\ L : (\mathbf{1}, \mathbf{2})_{-3}, \quad e^c : (\mathbf{1}, \mathbf{1})_6, \quad H : (\mathbf{1}, \mathbf{2})_3, \quad [\nu^c : (\mathbf{1}, \mathbf{1})_0]. \end{aligned}$$

The common kernel of these representations of $\tilde{G} = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)_y$ is exactly

$$K = \left\langle \left(e^{2\pi i/3} I_3, -I_2, e^{i\pi/3} \right) \right\rangle \cong \mathbb{Z}_6. \quad (73)$$

Consequently the represented matter packet is faithful precisely for $\tilde{G}/K$, which is the same global group selected intrinsically by the determinant seed in Theorem 4.19.

Proof. A common-kernel element must be central, so write it as $(\omega_3^a I_3, (-1)^b I_2, e^{i\theta})$. The charged singlet e^c gives $\theta = m\pi/3$ modulo 2π ; the lepton doublet gives $b \equiv m \pmod{2}$; and the colour antitriplet d^c gives $a \equiv m \pmod{3}$. Substitution in the remaining representations shows that precisely the six powers of the generator in (73) act trivially. The optional neutral singlet is trivial under all factors and adds no further restriction. $\square$

Remark 4.23 (Relative commutant versus physical gauge group). The two scalar summands in $M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}$ are components of the complex relative commutant; they are not two additional independent physical $\text{U}(1)$ gauge groups. The physical gauge group is obtained by preserving the represented determinant/incidence structure on the nonabelian multiplicity carriers and then requiring faithful action on the typed matter packet. Theorem 4.19 ties the central phases by $\det A \det B = 1$, while Proposition 4.22 removes the common $\mathbb{Z}_6$ kernel. Thus the passage from the relative commutant to G_{SM} is a stabilizer-and-faithfulness construction, not the operation of taking the full unitary group of $\mathcal{A}_{\text{int},r}^{\text{word}}$.

4.1.5 Anomaly-forced hypercharge

Anomaly cancellation is well known to constrain, and under appropriate representation assumptions essentially determine, Standard-Model charge assignments [31, 32]. In the noncommutative-geometric Standard Model it is closely tied to the unimodularity reduction of the gauge group [33]. Here the tensor types are already fixed by the preceding determinant/incidence construction; the theorem determines only the remaining central weight ratio.

Let the infinitesimal central action on $C \oplus W_2$ be

$$iaI_C \oplus ibI_{W_2}.$$

Define the tensor types

$$Q = C \otimes W_2, \quad u = \Lambda^2 C^*, \quad d = C, \quad L = W_2^*, \quad e = \Lambda^2 W_2^*, \quad H = W_2. \quad (74)$$

Their central weights are

$$(a + b, -2a, a, -b, -2b, b). \quad (75)$$

Theorem 4.24 (Anomaly-forced central weights). *Write right-handed fermions as left-handed conjugates when computing chiral anomalies. The local $\text{SU}(3)^2\text{U}(1)$, $\text{SU}(2)^2\text{U}(1)$, gravitational- $\text{U}(1)$, and cubic $\text{U}(1)$ anomalies vanish simultaneously if and only if*

$$\boxed{3a + 2b = 0.} \quad (76)$$

Up to overall reversal of the $\text{U}(1)$ orientation, the primitive integral solution is

$$(a, b) = (-2, 3).$$

With $y = 6Y$, the resulting physical weights are

$$\boxed{y(Q, u, d, L, e, H) = (1, 4, -2, -3, -6, 3).} \quad (77)$$

If a neutral right-handed singlet is retained, $y(\nu) = 0$.

Proof. After charge conjugation of the right-handed fields,

$$\begin{aligned} \mathcal{A}_{\text{SU}(3)^2 Y} &= \mathcal{A}_{\text{SU}(2)^2 Y} = \frac{1}{2}(3a + 2b), \\ \mathcal{A}_{\text{grav}^2 Y} &= 3(3a + 2b), \\ \mathcal{A}_{Y^3} &= 3(3a + 2b)(3a^2 + 2b^2). \end{aligned}$$

The pure colour anomaly cancels as $2 - 1 - 1$, and the weak doublet count is even. For a nontrivial abelian action $3a^2 + 2b^2 > 0$, so simultaneous local anomaly cancellation is equivalent to (76). The primitive coprime integral solution is $(-2, 3)$ up to sign; substitution gives (77). $\square$

4.1.6 Matter occurrence, generations, and the finite Dirac operator

In the usual almost-commutative Standard Model, three generations are encoded by tripling the finite fermion carrier [14, 11]. Other algebraic programmes have sought a structural origin for family replication, including complex-octonionic constructions [35], Clifford-ideal models of one generation [25], an intrinsic S_3 family action in a Clifford algebra [37], and the recent E_7 three-generation decomposition [38]; for a broader recent algebraic survey see [36]. The mechanism below is neither: the rank-three family carrier is a source-theoretic residual after spacetime shorting and remains subject to a positive lower-frame certificate.

The Clifford and internal packets must occur on one carrier. Let $J = P_+ - P_-$ be the protected pre-central grading and let $\sigma_0, \dots, \sigma_3$ be Hermitian Clifford axes generating the external matrix factor.

Definition 4.25 (Clifford matter margin). Define

$$p_{\text{Cl}} = \frac{1}{2 \dim \mathcal{H}} \sum_{\mu=0}^3 \|P_- \sigma_\mu P_+\|_{\text{HS}}^2. \quad (78)$$

Proposition 4.26 (Clifford occurrence test). *One has $p_{\text{Cl}} > 0$ exactly when at least one represented Clifford axis changes the protected grading. If $p_{\text{Cl}} = 0$, the grading reduces the external Clifford action and its residual action lies in the normal multiplicity commutant.*

Proof. Every summand in (78) is nonnegative. The sum vanishes exactly when $P_- \sigma_\mu P_+ = 0$ for every μ , and self-adjointness then also gives $P_+ \sigma_\mu P_- = 0$. Thus J commutes with the generated external algebra. $\square$

On the active residual of Theorem 4.1, the reversal-odd standard module W_- has rank three. We denote its supported complexification by $G_{\text{gen}} \cong \mathbb{C}^3$.

Proposition 4.27 (Endpoint generation carrier). *Suppose the restriction of the shorted Gram $G_{\text{int}|ST}$ to the reversal-odd endpoint block W_- in (48) is strictly positive. Then*

$$\dim_{\mathbb{C}} G_{\text{gen}} = 3. \quad (79)$$

The block $(W \otimes \text{sgn})_-$ is a distinct loop-provenance triplet and cannot be counted as a second generation source. On the explicit orthogonal active realization the hypothesis is automatic because $G_{\text{int}|ST} = \vartheta I_{12}$.

Proof. The endpoint and cycle projectors canonically separate the two odd irreducible summands in (48). The endpoint standard representation has complex dimension three. Strict positivity of the restricted shorted Gram prevents any endpoint direction from entering the source kernel, while the sign-twisted standard block lies in the distinct cycle projector. $\square$

Let A be a source-saturated self-adjoint full-cell generator on the common carrier. Its grading-odd part is

$$D_F = P_- A P_+ + P_+ A P_-. \quad (80)$$

Definition 4.28 (Odd provenance completeness). The branch is *odd-provenance complete* when every future-visible grading-odd action of the saturated full-cell carrier lies in the represented range of (80); equivalently, the orthogonal odd innovation Gram vanishes.

Proposition 4.29 (Canonical finite Dirac operator). *On an odd-provenance-complete saturated branch, D_F is the unique source-minimal self-adjoint grading-odd full-cell generator. Its typed contractions onto the modules in (74) give the finite Yukawa and optional Majorana residue blocks in one common generation frame.*

Proof. Source saturation makes the full-cell carrier reducing. The grading decomposition is orthogonal, so the odd component (80) is fixed by the represented generator. Vanishing of the orthogonal odd innovation removes any additional grading-odd source. Typed compression is then a deterministic operation on the same operator. $\square$

The weak multiplicity carrier W_2 supplies one primitive doublet. On the structural branch we require the primitive one-leg odd sector to be source complete; any extra weak-doublet copy would then produce a nonzero orthogonal source innovation and is therefore excluded by the same minimality criterion.

Theorem 4.30 (Structural Standard-Model classification from a landed multiplicity packet). *Let a finite source-complete joint packet satisfy the positive geometry short of Theorem 4.1. Then the following implications hold on the same represented carrier.*

- (i) *If one finite word depth satisfies the multiplicity census and internal landing hypotheses of Theorem 4.12, then the complete internal relative commutant is*

$$\boxed{M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}.$$

- (ii) *On that landed multiplicity packet, if the canonical fully alternating incidence projection of Proposition 4.17 is nonzero, then the physical determinant-preserving unitary group is*

$$\boxed{G_{\text{SM}} = S(U(3) \times U(2)) \cong \frac{SU(3) \times SU(2) \times U(1)}{\mathbb{Z}_6}.$$

- (iii) *For the represented tensor types in (74), local anomaly freedom is equivalent to $3a + 2b = 0$ and hence fixes the Standard-Model hypercharge ratios (77) up to overall orientation.*
- (iv) *If the shorted endpoint Gram on W_- is strictly positive, the internal endpoint source contains exactly one supported rank-three generation factor $G_{\text{gen}} \cong \mathbb{C}^3$, while the orthogonal sign-twisted triplet remains loop provenance.*
- (v) *If a represented grading-changing mixed word has $p_{\text{Cl}} > 0$ and the source-saturated full-cell generator is odd-provenance complete, its odd part is one source-minimal finite Dirac operator D_F ; the Yukawa/Majorana residue matrices are typed contractions of that operator in one common generation frame.*
- (vi) *If the primitive weak one-leg odd sector is source complete, the weak sector contains exactly one primitive doublet; any additional persistent copy produces a nonzero orthogonal source innovation.*

When all six clauses hold, the finite packet has the complete structural Standard-Model content claimed here. The nonabelian landing itself is not a dimension match: Theorems 4.3 and 4.4 give explicit no-go branches, while Theorem 4.13 gives the exact smaller internal algebra on every bridge/router failure.

Proof. Item (i) is Theorem 4.12. Item (ii) follows from Proposition 4.17 and Theorem 4.19. Item (iii) is Theorem 4.24; item (iv) is Proposition 4.27; item (v) is Propositions 4.26 and 4.29; and item (vi) is the source-minimality criterion for the primitive weak one-leg sector. $\square$

Corollary 4.31 (Historical typed-compilation criterion). *Let the relevant historical word hierarchy be flat. Suppose the grading, type, weak-leg and incidence source/target projections used to form the Store–Clifford and matter-incidence shadows are represented elements, or finite functional-calculus elements, of that same historical $\ast$ -algebra, and that every router or finite symmetry average entering the compression is word native in the sense of Definition 2.11. Then the complete mixed ancestry Gram of those typed shadows, its Schur ancestry residual, and the alternating determinant shadow are deterministic functions of the existing Grand-Tensor word Gram and multiplication table. No independent mixed cross-Hankel source row is required.*

Conversely, if a required typing projector or router is external to the historical algebra, Theorem 2.14 prevents its historical occurrence from being inferred from the old word moments alone.

Proof. The typed compressions are word-native linear shadows, so Theorem 2.12 and Corollary 2.13 reconstruct all ordinary-shadow and shadow-shadow blocks. Complete alternation on the already reconstructed typed Hom space is fixed linear postprocessing, so it introduces no additional physical source datum. The converse is exactly Theorem 2.14. $\square$

Remark 4.32 (Classification, obstruction, and selection). Theorem 4.30 does not project arbitrary data onto the Standard Model. Before the determinant or anomaly tests are even applied, the internal algebra is determined by a terminating multiplicity-side alternative. If the word hierarchy becomes flat with trivial multiplicity at most two, internal colour is impossible. If a three-dimensional colour-eligible multiplicity carrier occurs but the bridge vanishes, the colour block remains $M_2 \oplus \mathbb{C}$. If the weak router is absent, the weak multiplicity algebra remains $\mathbb{C}^2$. The four exact dimensions 9, 11, 13, 15 distinguish all such cases. On a fixed landed multiplicity census, full router saturation is generic by Corollary 4.14; microscopic occurrence of the required word rows remains a separate source-identification question.

Remark 4.33 (Occurrence certification versus microscopic source identification). The structural conclusions are finite system-bearing occurrence statements, not consequences of a classical opportunity record, an environment effect algebra, or a matching representation dimension. For a particular microscopic realization, the audit is finite and ordered: identify the actual reachable carrier and apply the module-recognition test when a regular word model is used; stop with Theorem 4.4 if a colour-eligible multiplicity never appears; test either a direct bridge or the reciprocal-return mass; verify the weak router and central separation; and only then apply the determinant and grading-changing incidence tests. When the required typing operations are historical and word native, Corollary 4.31 shows that their mixed ancestry data are already compiled by the Grand Tensor. The unresolved provenance question is then the occurrence of those typing operations themselves. The coarse-data and external-shadow no-go theorems explain why absent operations cannot be manufactured from record marginals or old word moments, while Theorem 2.5 gives an exact route from a complete operator-valued pullback to the represented action.

Scope statement 4.34 (Structural, not numerical). Theorem 4.30 does not determine the numerical entries of D_F , the nonzero pattern of every allowed residue, the flavour hierarchy, CKM or PMNS parameters, gauge couplings, the Higgs self-coupling or vacuum scale, renormalization-group running, matching, or the fermionic measure phase. These are coordinates on, or dynamical data over, the reconstructed structural packet rather than consequences of the commutant theorem.

The structural realization above identifies the desired finite multiplicity packet. We now give a basis-free positive test for exactness of the commutant and then show that the same duality is stable along controlled regulator refinement.

4.2 Finite certificates and regulator-stable duality

An exact finite commutant identity is useful only if it can be recognized without guessing a basis and if it remains stable under refinement. Both requirements are controlled by one positive operator.

4.2.1 The commutant Laplacian

Definition 4.35 (Commutant Laplacian). Let $c_1, \dots, c_s$ span a unital $*$ -algebra $\mathcal{C} \subseteq \mathcal{B}(\mathcal{H})$. On Hilbert-Schmidt operator space set

$$\mathcal{L} = \sum_{j=1}^s \text{ad}_{c_j}^* \text{ad}_{c_j}, \quad \text{ad}_c(X) = [c, X]. \quad (81)$$

Theorem 4.36 (Terminating commutant audit). *The operator $\mathcal{L}_C$ is positive and*

$$\boxed{\text{Ker } \mathcal{L}_C = C'}. \quad (82)$$

If $P_{C'}$ is Hilbert–Schmidt projection onto the commutant and $\lambda_C > 0$ is the least positive eigenvalue, then

$$\boxed{\|X - P_{C'}X\|_{\text{HS}}^2 \leq \lambda_C^{-1} \sum_j \|[c_j, X]\|_{\text{HS}}^2}. \quad (83)$$

Proof. For the Hilbert–Schmidt product, $\text{ad}_C^* = \text{ad}_{C'}^*$, hence

$$\langle X, \mathcal{L}_C X \rangle_{\text{HS}} = \sum_j \|[c_j, X]\|_{\text{HS}}^2.$$

The quadratic form vanishes exactly on simultaneous commutants. The positive spectral gap on $(C')^\perp$ gives (83). $\square$

4.2.2 The relative Howe Gram

Definition 4.37 (Relative Howe Gram). Let $\mathcal{M} \subseteq C^*(\mathbf{c})'$ be a proposed protected multiplicity algebra and let $E_1, \dots, E_r$ be a Hilbert–Schmidt orthonormal basis of $\mathcal{M}^\perp$. Define

$$\boxed{\mathbb{G}_{\text{Howe}}(\mathbf{c} \mid \mathcal{M})_{ab} = \sum_j \langle [c_j, E_a], [c_j, E_b] \rangle_{\text{HS}}}. \quad (84)$$

Theorem 4.38 (Positive certificate for the exact multiplicity algebra). *Under Definition 4.37,*

$$\boxed{\mathbb{G}_{\text{Howe}}(\mathbf{c} \mid \mathcal{M}) \succ 0 \iff C^*(\mathbf{c})' = \mathcal{M}}. \quad (85)$$

On this branch the least eigenvalue of the Howe Gram is the first positive eigenvalue of the commutant Laplacian.

Proof. The matrix in (84) is the restriction of $\mathcal{L}_{C^*(\mathbf{c})}$ to $\mathcal{M}^\perp$. Since $\mathcal{M}$ is already contained in the kernel, the restriction is positive definite exactly when the kernel has no additional vector. $\square$

Thus the proposed generation/flavour algebra is not accepted by dimensional matching. It is certified by one positive finite matrix whose null directions are exactly unaccounted commutants.

4.2.3 Howe discriminant and generic exactness

The positive certificate also determines the symmetry-enhancement locus of a parameter family. Let $\Theta \subset \mathbb{R}^p$ be an open parameter domain, keep a fixed represented protected algebra $\mathcal{M} \subseteq \mathcal{B}(\mathcal{H})$, and assume $[c_j(\theta), \mathcal{M}] = 0$ for every j and θ . On $\mathcal{K} := \mathcal{M}^\perp$ define the stacked commutator map

$$D(\theta)X = ([c_1(\theta), X], \dots, [c_s(\theta), X]), \quad G(\theta) = D(\theta)^* D(\theta).$$

In an orthonormal basis of $\mathcal{K}$, $G(\theta)$ is the relative Howe Gram. Put $r = \dim_{\mathbb{C}} \mathcal{K}$ and

$$\Delta_{\mathcal{M}}(\theta) = \det G(\theta). \quad (86)$$

Theorem 4.39 (Howe enhancement discriminant and generic rigidity). *Assume every matrix entry of every $c_j(\theta)$ is a complex polynomial in the real coordinates θ , of degree at most d . Then $\Delta_{\mathcal{M}}$ is a nonnegative real polynomial of degree at most $2dr$ and admits the Cauchy–Binet representation*

$$\Delta_{\mathcal{M}}(\theta) = \sum_{|I|=r} |\det D_I(\theta)|^2. \quad (87)$$

Moreover,

$$\{\theta : \Delta_{\mathcal{M}}(\theta) = 0\} = \{\theta : C^*(\mathbf{c}(\theta))' \supsetneq \mathcal{M}\}. \quad (88)$$

If the proposed commutant is exact at one parameter value, then $\Delta_{\mathcal{M}}$ is not the zero polynomial. Hence its enhancement locus has empty interior and Lebesgue measure zero, while exact duality holds on an open dense full-measure subset of every open admissible parameter region. The same open-dense conclusion holds for a real-analytic family on each connected analytic component containing one exact point.

Proof. The entries of $D(\theta)$ are polynomial of degree at most d . Cauchy–Binet applied to D^*D gives (87); each squared modulus is a sum of squares of real polynomials, proving nonnegativity and the degree bound. Since $G \succeq 0$, its determinant is positive exactly when $G \succ 0$, which is equivalent to $C^*(\mathbf{c}(\theta))' = \mathcal{M}$ by Theorem 4.38. A nonzero real polynomial has a zero set of Lebesgue measure zero and empty interior; continuity makes the complement open. For a connected real-analytic family, the identity theorem gives the corresponding empty-interior conclusion once the determinant is nonzero at one point. $\square$

Corollary 4.40 (Determinantal nullity filtration). *For $1 \leq k \leq r$, the locus on which the commutant acquires at least k additional complex dimensions is the common zero set of all $(r-k+1) \times (r-k+1)$ minors of $D(\theta)$. Additional commutant dimension is therefore upper semicontinuous in every continuous family.*

Remark 4.41 (Genericity is relative to the admissible source class). Theorem 4.39 does not assert that microscopic Renewal dynamics generically enters a chosen structural Standard-Model family. It states that, once a fixed protected algebra and an admissible polynomial or analytic source family are specified, accidental enlargement of the commutant is confined to the corresponding enhancement locus.

4.2.4 A finite anchor locks the commutant

Theorem 4.42 (Finite-anchor stability). *Transport a cutoff family to one common finite screened carrier and write $\mathbf{c}_X = (c_{1,X}, \dots, c_{s,X})$. Suppose a fixed protected algebra $\mathcal{M}$ commutes with every tuple. Set*

$$C_X = \left(\sum_j \|c_{j,X}\|_{\text{op}}^2 \right)^{1/2}, \quad \varepsilon_{X,Y} = \left(\sum_j \|c_{j,X} - c_{j,Y}\|_{\text{op}}^2 \right)^{1/2}.$$

If at one cutoff X_0

$$\mathbb{G}_{\text{Howe}}(\mathbf{c}_{X_0} \mid \mathcal{M}) \succeq \gamma_0 I, \quad \gamma_0 > 0,$$

and for every later X

$$4(C_X + C_{X_0})\varepsilon_{X,X_0} < \gamma_0, \quad (89)$$

then

$$C^*(\mathbf{c}_X)' = \mathcal{M}, \quad \lambda_{\min}^+(\mathcal{L}_{C^*(\mathbf{c}_X)}) \geq \gamma_0 - 4(C_X + C_{X_0})\varepsilon_{X,X_0}. \quad (90)$$

Proof. The commutator derivations satisfy

$$\|\partial_{\mathbf{c}_X} - \partial_{\mathbf{c}_Y}\|_{2 \rightarrow 2} \leq 2\varepsilon_{X,Y}, \quad \|\partial_{\mathbf{c}_X}\|_{2 \rightarrow 2} \leq 2C_X.$$

Using $A^*A - B^*B = A^*(A - B) + (A^* - B^*)B$ gives

$$\|\mathcal{L}_X - \mathcal{L}_Y\| \leq 4(C_X + C_Y)\varepsilon_{X,Y}.$$

The anchor Laplacian is at least $\gamma_0 I$ on $\mathcal{M}^\perp$. The perturbation estimate gives (90). $\square$

4.2.5 Compact-Mosco cofinal duality

The perturbative anchor theorem controls one finite neighborhood in generator norm. A distinct nonperturbative mechanism is available for a genuine cofinal family. We formulate it on transported screened Hilbert spaces and derive norm-resolvent and spectral convergence from the form hypotheses rather than assuming them. This is the compact form of the Mosco convergence mechanism; see Kuwae-Shioya [42]. The specialized argument required here is recorded in Appendix F.1.

Let $\mathcal{H}$ be a separable Hilbert space and let $J_X : \mathcal{H}_X \rightarrow \mathcal{H}$ be isometric embeddings of the screened Hilbert-Schmidt operator spaces, with $P_X = J_X J_X^*$ the corresponding orthogonal projections. The limiting space is embedded by J_∞ , and we assume the transport is compatible in the sense that $P_X \rightarrow P_\infty$ strongly on $\mathcal{H}$. For a densely defined closed nonnegative form q_X on $\mathcal{H}_X$, denote its self-adjoint operator by $\mathcal{L}_X$ and its transported resolvent by

$$R_X := J_X(\mathcal{L}_X + I)^{-1} J_X^* \in \mathcal{B}(\mathcal{H}). \quad (91)$$

Definition 4.43 (Transported Mosco convergence). The forms q_X converge to q_∞ in the transported Mosco sense when:

(M1) whenever $J_X A_X \rightharpoonup B$ weakly in $\mathcal{H}$, the strong convergence $P_X \rightarrow P_\infty$ gives $B \in J_\infty \mathcal{H}_\infty$; writing $B = J_\infty A$,

$$q_\infty(A) \leq \liminf_X q_X(A_X);$$

(M2) for every $A \in D(q_\infty)$ there are $A_X \in D(q_X)$ such that $J_X A_X \rightarrow J_\infty A$ strongly and

$$q_X(A_X) \rightarrow q_\infty(A).$$

Definition 4.44 (Asymptotic form compactness). The family (q_X) is asymptotically compact if every sequence $A_X \in D(q_X)$ satisfying

$$\sup_X (\|A_X\|^2 + q_X(A_X)) < \infty \quad (92)$$

has a subsequence for which $J_X A_X$ converges strongly in $\mathcal{H}$. The same requirement is imposed on the limiting form. In particular, $(\mathcal{L}_\infty + I)^{-1}$ is compact on the limiting screened space.

Proposition 4.45 (Compact Mosco convergence implies norm-resolvent convergence). *If $q_X \rightarrow q_\infty$ in the sense of Definition 4.43 and the family is asymptotically compact in the sense of Definition 4.44, then*

$$\|R_X - R_\infty\|_{\text{op}} \rightarrow 0. \quad (93)$$

Consequently, if $0 \leq \lambda_{1,X} \leq \lambda_{2,X} \leq \dots$ denote the discrete screened eigenvalues of $\mathcal{L}_X$, repeated with multiplicity, then for every fixed index in the limiting discrete spectrum,

$$\lambda_{k,X} \rightarrow \lambda_{k,\infty}. \quad (94)$$

If an isolated spectral cluster of $\mathcal{L}_\infty$ is separated by a positive gap, the corresponding transported Riesz projections converge in operator norm and their ranks are eventually constant.

Proof. Transported Mosco convergence gives strong convergence of the resolvents. The resolvent equation bounds $\|u_X\|^2 + q_X(u_X)$ uniformly for $u_X = (\mathcal{L}_X + I)^{-1}f_X$ whenever (f_X) is bounded. Asymptotic form compactness therefore makes the transported resolvents collectively compact. Because they are self-adjoint, collective compactness together with strong convergence upgrades the convergence to operator norm. Norm convergence of compact self-adjoint resolvents gives (94); the Riesz-projection statement follows from the contour resolvent identity. A detailed proof, including the varying-space bookkeeping, is given in Appendix F.1. $\square$

Theorem 4.46 (Compact-Mosco cofinal spacetime–gauge duality). *Let*

$$q_X(A) = \sum_j \| [c_{j,X}, A] \|_{\text{HS}}^2 \quad (95)$$

be the closed commutator forms of a cofinal family of screened support-polar or multiplicity-quiver generators, with associated commutant Laplacians $\mathcal{L}_X$. Assume:

- (H1) $q_X \rightarrow q_\infty$ in transported Mosco sense;
- (H2) (q_X) is asymptotically form compact;
- (H3) for every X there is a protected multiplicity algebra $\mathcal{M}_X \subseteq \text{Ker } \mathcal{L}_X$ of the same finite complex dimension m ;
- (H4) the limiting commutant contains no unprotected direction, $\text{Ker } \mathcal{L}_\infty = \mathcal{M}_\infty$ and $\dim_{\mathbb{C}} \mathcal{M}_\infty = m$.

Then the limiting operator has compact resolvent and

$$\gamma_\infty := \lambda_{m+1}(\mathcal{L}_\infty) > 0. \quad (96)$$

For every sufficiently late cutoff,

$$\boxed{\text{Ker } \mathcal{L}_X = \mathcal{M}_X, \quad C^*(\mathbf{c}_X)' = \mathcal{M}_X,} \quad (97)$$

and

$$\boxed{\lambda_{\min}^+(\mathcal{L}_X) = \lambda_{m+1}(\mathcal{L}_X) \longrightarrow \gamma_\infty.} \quad (98)$$

If Π_X denotes the transported spectral projection onto $\text{Ker } \mathcal{L}_X$, then

$$\boxed{\|\Pi_X - \Pi_\infty\|_{\text{op}} \longrightarrow 0.} \quad (99)$$

Consequently, for every $0 < \gamma_ < \gamma_\infty$ and all sufficiently late X ,*

$$\boxed{\sum_j \| [c_{j,X}, A] \|_{\text{HS}}^2 \geq \gamma_* \|(I - P_{\mathcal{M}_X})A\|_{\text{HS}}^2.} \quad (100)$$

Proof. Proposition 4.45 gives norm convergence of the transported resolvents and hence convergence of every fixed low eigenvalue and of the isolated zero-cluster projection. Compactness makes the limiting spectrum discrete near zero. By (H4), zero has multiplicity exactly m , so (96) holds. By (H3), every finite cutoff has at least m exact zero modes. Norm convergence of the Riesz projection onto the isolated eigenvalue 1 of $(\mathcal{L}_\infty + I)^{-1}$ implies that the corresponding finite-cutoff spectral subspace has dimension exactly m for all sufficiently late X . Hence no additional zero mode is possible and $\text{Ker } \mathcal{L}_X = \mathcal{M}_X$. Eigenvalue convergence gives (98), projection convergence gives (99), and the spectral theorem gives (100). $\square$

Remark 4.47 (Two distinct stability mechanisms). Theorems 4.42 and 4.46 are logically independent. The former is a quantitative perturbative statement in generator norm around one positive finite anchor. The latter is nonperturbative: it allows the carrier and form domains to vary and derives spectral stability from Mosco convergence plus compactness. Either mechanism can lock the same finite commutant; neither is a substitute for occurrence of the underlying system-bearing generators.

Remark 4.48 (Independence of the screen construction). Theorem 4.46 depends only on the transported form hypotheses (H1)–(H4). Any regulator family satisfying them has the stated stable commutant, regardless of the microscopic origin of its compactness. Ref. [2] supplies one Renewal-Geometric mechanism for the required screened form control.

Remark 4.49 (What the cofinal theorem does and does not say). Theorem 4.46 stabilizes the algebraic duality and the protected multiplicity spectral subspace. It does not by itself construct an interacting four-dimensional quantum field theory, determine running couplings, remove fermionic determinant-line phases, or construct a continuum measure.

The commutant duality and its coercive stability are algebraic statements. We finally place the same internal sector on the common variational carrier of the published spacetime construction and isolate the source-minimal gravity–matter coupling.

4.3 The common gravity–internal action

The duality theorem is algebraic. We now formulate the source-minimal variational coupling abstractly on a common gravitational/internal carrier. The spacetime companion paper [?] supplies the Renewal-Geometric realization of the gravitational side of this common finite carrier. The purpose of the subsection is to establish compatibility of the dual pair with one gravity–matter action, not to reproduce the separate functional-analytic problem of constructing its full continuum limit.

4.3.1 The internal source after gravitational shorting

Let

$$S_g : E_g \rightarrow \mathcal{H}_{\text{cyl}}, \quad S_i : E_i \rightarrow \mathcal{H}_{\text{cyl}}$$

be the source syntheses of the complete gravitational and internal physical variation banks on one same-history cylinder. Assume the gravitational bank has already been reduced to its faithful quotient and put

$$G_g = S_g^* S_g, \quad G_i = S_i^* S_i, \quad C = S_i^* S_g.$$

Theorem 4.50 (Universal coupled source quotient). *Define*

$$\boxed{G_{i|g} = G_i - C G_g^{-1} C^* = S_i^* (I - P_g) S_i.} \quad (101)$$

Then

$$\boxed{\begin{pmatrix} G_g & C^* \\ C & G_i \end{pmatrix} = \begin{pmatrix} I & 0 \\ C G_g^{-1} & I \end{pmatrix} \begin{pmatrix} G_g & 0 \\ 0 & G_{i|g} \end{pmatrix} \begin{pmatrix} I & G_g^{-1} C^* \\ 0 & I \end{pmatrix}.} \quad (102)$$

Consequently:

- (i) *the joint variation carrier is faithful exactly when $G_{i|g} \succ 0$;*
- (ii) *the number of genuinely new internal source directions is $\text{rank } G_{i|g}$;*

(iii) *the source-minimal internal variation space is*

$$E_{i|g}^{\min} = E_i / \text{Ker } G_{i|g}.$$

Proof. The first equality in (101) is the Gram of $(I - P_g)S_i$. Direct block multiplication gives (102); the remaining statements are the standard Schur-complement rank and positivity consequences. $\square$

Thus “adding the Standard Model” does not mean appending every internal variation as a new primitive. Only the component orthogonal to the already represented gravitational source counts as genuinely new.

4.3.2 Finite common-action representative

The duality theorem does not require a gauge–matter variational action. For completeness, Appendix E constructs one explicit finite gauge–Higgs–Dirac representative on the reconstructed spacetime complex and proves its exact gauge, Ward, BRST, stress, and stationary vacuum identities. Those formulas provide a concrete realization for the compatibility statement below, but they are not used in the proof of the finite double-centralizer theorem or in the structural Standard-Model reconstruction.

4.3.3 Classical compatibility of the stabilized dual branch

The finite common-action construction is sufficient for the purpose of the present paper. To state precisely what it implies, let $\mathcal{S}_{\text{SM},X}^L$ denote the Lorentzian signed counterpart of (109), obtained from the reconstructed temporal Hodge signs. We do not attempt here to prove the independent geometric and PDE estimates required to construct a general Einstein–Standard-Model continuum from microscopic renewal data. Instead we record the exact compatibility implication that any such certified continuum realization must satisfy.

Proposition 4.51 (Classical compatibility of the stabilized dual branch). *Let a cofinal family satisfy the coercive multiplicity duality of Theorem 4.46. Suppose, in addition, that the gravitational and internal fields are transported to one common determining test core such that:*

- (C1) *the gravitational first-variation functionals converge to the Einstein–Hilbert/Palatini first variation with cosmological term;*
- (C2) *the first variations of $\mathcal{S}_{\text{SM},X}^L$ converge to the classical Yang–Mills–Higgs–Dirac–Yukawa first variation, including the metric variation defining $T_{\mu\nu}^{\text{SM}}$;*
- (C3) *for a determining transported gravity–matter variation bank,*

$$D(\mathcal{S}_{g,X} + \mathcal{S}_{\text{SM},X}^L)[\dot{z}_X] \longrightarrow 0.$$

Then every limiting field configuration satisfies the classical matter Euler equations on that test core and

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{SM}}} \tag{103}$$

distributionally, with κ fixed by the relative normalization of the gravity and matter terms in the common action. Along the same cofinal family, the limiting matter fields carry the stabilized multiplicity algebra of Theorem 4.46; no additional unprotected flavour/generation direction is created by the handoff.

Proof. Take first a pure compactly supported matter variation in the determining bank. By hypothesis (C2), the finite matter first variations converge to the classical Yang–Mills–Higgs–Dirac–Yukawa first variation, whereas the gravity variation vanishes. Passing to the limit in (C3) therefore gives the matter Euler equations.

Next take a pure compactly supported metric variation $k^{\mu\nu}$. With fixed nonzero normalizations c_g and c_m for the gravitational and matter parts, hypotheses (C1)–(C2) give

$$c_g \int (G_{\mu\nu} + \Lambda g_{\mu\nu}) k^{\mu\nu} d\text{vol}_g - \frac{c_m}{2} \int T_{\mu\nu}^{\text{SM}} k^{\mu\nu} d\text{vol}_g = 0,$$

up to the fixed stress convention used in Theorem E.1. Absorbing that convention into κ and using that the metric test bank is determining yields (103) as a distributional identity.

Finally, Theorem 4.46 identifies the multiplicity kernel for all sufficiently late cutoffs and gives a uniform positive commutant gap. Hence the matter packet entering the limiting action is the same stabilized structural packet reconstructed by the finite duality theorem, rather than a larger multiplicity sector generated by refinement. $\square$

Scope statement 4.52 (Continuum analysis is separate from the duality theorem). Proposition 4.51 is a compatibility statement, not a construction of a general cofinal Einstein–Standard-Model solution. Establishing the hypotheses above from microscopic renewal dynamics requires a separate analytic theory of geometric reconstruction, nonlinear stress compactness, constraint propagation, boundary/domain control, and strong coupled-source convergence. Those problems are logically downstream of the finite commutant duality and are not used anywhere in its proof or in the structural Standard-Model reconstruction.

5 Discussion and outlook

The main result fixes the precise meaning of “spacetime–gauge duality” in this framework. The duality is not an identification of the Lorentzian Clifford algebra with the Standard-Model gauge algebra, nor an assertion that one is the ordinary commutant of the other. The factor obstruction shows that such a statement is algebraically impossible on the relevant reducible typed carrier. The exact dual pair is instead formed by the *joint typed action*—spacetime, internal representation, and physical incidence together—and the matter multiplicity that remains invisible to that complete action.

This distinction matters physically. A gauge representation tells us how a field transforms, but it does not by itself determine how many equivalent copies of that representation occur. In the present construction that multiplicity is not an unrelated label added after the gauge sector has been chosen. It is the commutant of the complete represented action. Conversely, once the protected multiplicity algebra is known, finite bicommutant closure recovers the joint action algebra. The two sides are therefore complementary descriptions of one represented finite packet.

Corollary 3.12 makes this complementarity explicit. The common carrier splits into action–multiplicity tensor blocks; inside each central sector the joint action is a full matrix algebra on one factor and the residual multiplicity is a full matrix algebra on the other. Both sides have the same centre. Thus the duality is stronger than equality of two abstract dimensions: it is a reciprocal finite factorization of the represented packet, canonical up to simultaneous unitary equivalence.

The singular theory is important for the same reason. Yukawa or incidence maps can have zero modes, unequal source and target dimensions, or noncommuting supports. None of these cases requires inserting an inverse matrix or completing a missing edge by hand. The polar support

algebra is defined directly from the occurring maps, and its commutant is exact. When the support projections themselves carry nontrivial geometry, the canonical multiplicity quiver records it. The surviving flavour algebra is then the endomorphism algebra of that quiver.

The incidence bank itself has an intrinsic minimality theory. The grouped commutator constraints define the submodular rank function (40); deleting an edge is harmless exactly when that rank is unchanged, and every finite exact packet contains an inclusion-minimal incidence skeleton. Thus “physical incidence is essential” is a terminating rank statement rather than a graphical heuristic.

The source-complete requirement is likewise operational rather than terminological. Theorem 2.4 shows that even a complete classical record history and matching coarse Choi data can leave the acted-on system algebra ambiguous. Conversely, the operator-valued complementary pullback of Theorem 2.5 recovers the system action exactly when an occurring identity pivot is present. The duality theorem is therefore unconditional once the system-bearing packet is specified, while microscopic provenance is a distinct source-identification problem.

The positive Howe Gram turns this structural statement into a finite certificate. Instead of asserting that a proposed generation or flavour algebra is the full commutant, one computes a positive matrix on its orthogonal complement. A zero eigenvector is an explicit missing symmetry; a positive least eigenvalue is a rigidity margin. For polynomial or analytic source families, Theorem 4.39 sharpens this statement: commutant enhancement lies on a determinantal locus and exact duality is generic within every admissible component containing one exact point. Cofinal stability is likewise a conclusion of the operator theory: transported Mosco convergence and asymptotic compactness yield norm-resolvent convergence, convergence of the protected commutant projection, and a uniform late-cutoff coercive gap.

The Standard-Model result should be read at the same structural level. The active residual does not obtain $M_3(\mathbb{C})$ and $M_2(\mathbb{C})$ by noticing matrix blocks of dimensions three and two in the tetrahedral representation: those direct Burnside blocks are external. Internal factors live on multiplicity spaces, and Theorem 2.8 further distinguishes kinematic invariant endomorphisms of a word space from operations induced by the historical system algebra. At the populated categorical degree-one census internal colour is impossible; if the physical word hierarchy becomes flat before an external-trivial multiplicity of three appears, colour is ruled out at every later depth. Once the multiplicity census $(3, 2, 1, 1)$ is present, the nonabelian landing requires only a colour bridge and a noncommuting weak router; on a reciprocal common-mediator branch the bridge is itself forced by the positive return mass (62). A separate central separator rules out the only remaining 14-dimensional scalar-locked subdirect product before the full $M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}$ commutant is asserted. Under independent central supports the bridge/router deficits are the exact 9, 11, 13, 15 alternatives already displayed. Only after this internal landing is the determinant incidence tested. Its stabilizer is $S(U(3) \times U(2))$; a nonzero coherent determinant cross mass gives the same group operationally, while the sharp arity bound shows that this character cannot be manufactured by lower-order defining tensor ports. Anomaly freedom then fixes the remaining central weight ratio.

This provides a useful contrast with the almost-commutative formulation of noncommutative geometry. There, the Standard Model is encoded by a finite spectral geometry combined with a continuum spin manifold, and the spectral action produces the coupled field theory [7, 8, 9, 10, 14]. Krajewski classification and the later Hodge/Morita programme make the finite incidence and commutant structures of that input highly nontrivial [12, 17, 18]. Here the finite spectral packet and Lorentzian spacetime are themselves downstream outputs of the same predictive Grand Tensor. The present duality is therefore not a replacement of the almost-commutative product by another chosen product. It is a statement about the mutual commutants that remain after both downstream

sectors are realized on one common carrier.

The generation result is perhaps the most delicate structural consequence. Three generations do not follow merely because the spacetime construction contains a three-dimensional module. The geometric copy is removed first. What remains on the internal categorical contrast is a distinct reversal-odd standard module of rank three. Its orthogonal sign-twisted triplet is kept as loop provenance. Thus the finite source calculation contains an exact mechanism that distinguishes the generation triplet from a second geometric or cycle copy. Whether the physical flavour matrices select the observed hierarchy and mixing inside that triplet is a separate question.

The same separation applies to the finite Dirac operator. The framework does not derive five unrelated Yukawa matrices and then place them into a larger operator. On the source-saturated branch, one grading-odd full-cell generator is reconstructed first; the residue matrices are typed contractions of that one operator in one generation frame. Representation theory determines the allowed slots, not their numerical values. Flavour breaking, masses, mixing angles, and CP phases therefore remain genuine dynamical information.

The common-action Schur quotient adds a final compatibility point without enlarging the duality theorem. Internal variations count as genuinely new source directions only through the residual $G_{i|g} = S_i^*(I - P_g)S_i$. This is the variational analogue of the algebraic construction: in both cases the internal sector is defined relative to the already represented spacetime source rather than appended by direct sum. The explicit finite gauge–matter action in Appendix E shows that the stabilized dual packet is compatible with later coupled dynamics; no Einstein–Standard-Model continuum limit is needed for the finite mutual-commutant result.

Several open problems remain. The first is microscopic source identification and dynamical selection, but the unresolved interface is narrower than a list of independent mixed Gram entries. Once the historical word algebra and its multiplication table are known, every mixed ancestry quantity associated with a word-native typing operation is already an inserted Grand-Tensor moment. The source-level question is therefore whether the grading, type, incidence and router operations used by the structural compiler themselves occur in the historical flat algebra, and whether an actual determinant-bearing typed port is present. If so, the ancestry and determinant tests become finite table-native computations; if not, Theorem 2.14 shows that the missing operation cannot be inferred from the old moments alone. The present theorems do not assert that generic microscopic dynamics selects this terminal branch. A second problem is parameter selection. Nothing here fixes the observed gauge couplings, Higgs parameters, Yukawa spectra, or mixing matrices. Finally, the full classical Einstein–Standard-Model continuum is analytically distinct: deriving the transported Mosco and asymptotic-compactness hypotheses, together with the convergence assumptions of Proposition 4.51, from a single microscopic cofinal family requires geometric reconstruction, nonlinear stress compactness, constraint propagation, and coupled-source estimates that are not consequences of the commutant theorem. Beyond that lies the quantum problem, including the determinant/Pfaffian phase, renormalization, and the construction of an interacting four-dimensional continuum measure.

Within these limits, the logical research hierarchy is naturally staged. The spacetime paper reconstructs Lorentzian geometry and gravitational dynamics, and the spectral paper reconstructs a Connes-type effective geometry and characterizes information lost under spectralization. The present paper uses those works to supply the independently reconstructed Lorentzian and spectral inputs, while its finite duality and structural branch are autonomous consequences of the explicit joint-packet and occurrence hypotheses stated here. A separate continuum analysis may then ask when the finite common gravity–matter action converges to the classical Einstein–Standard-Model

system. The logical content of the present paper ends one step earlier:

$$\begin{array}{l}
\text{Renewal Grand Tensor} \longrightarrow \text{Lorentzian + spectral common carrier} \\
\longrightarrow \text{joint typed action} \\
\overset{\text{commutant}}{\longleftrightarrow} \text{matter multiplicity} \\
\longrightarrow \text{structural Standard Model.}
\end{array} \tag{104}$$

The central conclusion is therefore not that internal gauge symmetry is spacetime in disguise. Rather, spacetime action and matter multiplicity are complementary operator-algebraic faces of one predictive carrier once the physical incidence between them is retained. The finite common-action construction shows that this dual pair is compatible with later gravity–matter dynamics, while the analytic continuum construction is deliberately left to a separate work.

Data and formalization availability

The paper is mathematical and uses no external empirical dataset. The finite algebraic statements are compatible with the formalization programme at <https://github.com/AI-MathPhys/renewal-geometry>. Publication metadata for the two companion Renewal Geometry papers should be updated in the bibliography when final journal details are available.

References

- [1] A. Pélissier, Renewal Geometry and the emergence of Lorentzian spacetime, HAL preprint hal-05744772, 2026. <https://hal.science/hal-05744772>
- [2] A. Pélissier, From predictive dynamics to spectral geometry, 2026.
- [3] S. Coleman and J. Mandula, All possible symmetries of the S matrix, *Physical Review* **159** (1967), 1251–1256. doi:10.1103/PhysRev.159.1251.
- [4] A. Connes and J. Lott, Particle models and noncommutative geometry, *Nuclear Physics B – Proceedings Supplements* **18** (1991), 29–47. doi:10.1016/0920-5632(91)90120-4.
- [5] A. Connes, *Noncommutative Geometry*, Academic Press, San Diego, 1994.
- [6] A. Connes, Noncommutative geometry and reality, *Journal of Mathematical Physics* **36** (1995), no. 11, 6194–6231. doi:10.1063/1.531241.
- [7] A. Connes, Gravity coupled with matter and the foundation of non-commutative geometry, *Communications in Mathematical Physics* **182** (1996), no. 1, 155–176. doi:10.1007/BF02506388.
- [8] A. H. Chamseddine and A. Connes, The spectral action principle, *Communications in Mathematical Physics* **186** (1997), 731–750. doi:10.1007/s002200050126.
- [9] A. H. Chamseddine, A. Connes, and M. Marcolli, Gravity and the standard model with neutrino mixing, *Advances in Theoretical and Mathematical Physics* **11** (2007), no. 6, 991–1089. doi:10.4310/ATMP.2007.v11.n6.a3.
- [10] A. H. Chamseddine and A. Connes, Why the Standard Model, *Journal of Geometry and Physics* **58** (2008), no. 1, 38–47. doi:10.1016/j.geomphys.2007.09.011.

- [11] W. D. van Suijlekom, *Noncommutative Geometry and Particle Physics*, 2nd ed., Mathematical Physics Studies, Springer, 2025.
- [12] T. Krajewski, Classification of finite spectral triples, *Journal of Geometry and Physics* **28** (1998), 1–30. doi:10.1016/S0393-0440(97)00068-5.
- [13] B. Iochum, T. Schücker, and C. Stephan, On a classification of irreducible almost commutative geometries, *Journal of Mathematical Physics* **45** (2004), 5003–5041. doi:10.1063/1.1811372.
- [14] K. van den Dungen and W. D. van Suijlekom, Particle physics from almost commutative spacetimes, *Reviews in Mathematical Physics* **24** (2012), 1230004. doi:10.1142/S0129055X1230004X.
- [15] A. H. Chamseddine, A. Connes, and W. D. van Suijlekom, Beyond the spectral Standard Model: emergence of Pati–Salam unification, *Journal of High Energy Physics* **2013** (2013), no. 11, 132. doi:10.1007/JHEP11(2013)132.
- [16] L. Boyle and S. Farnsworth, A new algebraic structure in the standard model of particle physics, *Journal of High Energy Physics* **2018** (2018), no. 6, 071. doi:10.1007/JHEP06(2018)071.
- [17] F. D’Andrea and L. Dabrowski, The Standard Model in noncommutative geometry and Morita equivalence, *Journal of Noncommutative Geometry* **10** (2016), 551–578. doi:10.4171/JNCG/242.
- [18] L. Dabrowski, F. D’Andrea, and A. Sitarz, The Standard Model in noncommutative geometry: fundamental fermions as internal forms, *Letters in Mathematical Physics* **108** (2018), 1323–1340. doi:10.1007/s11005-017-1036-x.
- [19] L. Dabrowski and A. Sitarz, Fermion masses, mass-mixing and the almost commutative geometry of the Standard Model, *Journal of High Energy Physics* **2019** (2019), no. 2, 68. doi:10.1007/JHEP02(2019)068.
- [20] J. Bhowmick, F. D’Andrea, B. Das, and L. Dabrowski, Quantum gauge symmetries in noncommutative geometry, *Journal of Noncommutative Geometry* **8** (2014), 433–471. doi:10.4171/JNCG/161.
- [21] J. W. Barrett, A Lorentzian version of the non-commutative geometry of the standard model of particle physics, *Journal of Mathematical Physics* **48** (2007), 012303. doi:10.1063/1.2408400.
- [22] A. Devastato, S. Farnsworth, F. Lizzi, and P. Martinetti, Lorentz signature and twisted spectral triples, *Journal of High Energy Physics* **2018** (2018), no. 3, 89. doi:10.1007/JHEP03(2018)089.
- [23] G. Trayling and W. E. Baylis, A geometric basis for the standard-model gauge group, *Journal of Physics A: Mathematical and General* **34** (2001), 3309–3324. doi:10.1088/0305-4470/34/15/309.
- [24] C. Daviau, Gauge group of the Standard Model in $Cl_{1,5}$, *Advances in Applied Clifford Algebras* **27** (2017), 279–290. doi:10.1007/s00006-015-0566-5.
- [25] N. G. Gresnigt, The Standard Model particle content with complete gauge symmetries from minimal ideals of two Clifford algebras, *European Physical Journal C* **80** (2020), 583. doi:10.1140/epjc/s10052-020-8141-1.
- [26] S. Doplicher and J. E. Roberts, A new duality theory for compact groups, *Inventiones Mathematicae* **98** (1989), 157–218. doi:10.1007/BF01388849.

- [27] S. Doplicher and J. E. Roberts, Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics, *Communications in Mathematical Physics* **131** (1990), 51–107. doi:10.1007/BF02097680.
- [28] I. Todorov, Gauge symmetry and Howe duality in 4D conformal field theory models, *Advances in Mathematical Physics* **2010** (2010), Article ID 509538, 12 pages. doi:10.1155/2010/509538.
- [29] J. Hucks, Global structure of the standard model, anomalies, and charge quantization, *Physical Review D* **43** (1991), 2709–2717. doi:10.1103/PhysRevD.43.2709.
- [30] D. Tong, Line operators in the Standard Model, *Journal of High Energy Physics* **2017** (2017), no. 7, 104. doi:10.1007/JHEP07(2017)104.
- [31] C. Q. Geng and R. E. Marshak, Uniqueness of quark and lepton representations in the Standard Model from the anomalies viewpoint, *Physical Review D* **39** (1989), 693–696. doi:10.1103/PhysRevD.39.693.
- [32] J. A. Minahan, P. Ramond, and R. C. Warner, Comment on anomaly cancellation in the Standard Model, *Physical Review D* **41** (1990), 715–716. doi:10.1103/PhysRevD.41.715.
- [33] E. Alvarez, J. M. Gracia-Bondía, and C. P. Martín, Anomaly cancellation and the gauge group of the Standard Model in NCG, *Physics Letters B* **364** (1995), 33–40. doi:10.1016/0370-2693(95)01051-3.
- [34] K. Krasnov, SO(9) characterization of the Standard Model gauge group, *Journal of Mathematical Physics* **62** (2021), 021703. doi:10.1063/5.0039941.
- [35] C. Furey, Three generations, two unbroken gauge symmetries, and one eight-dimensional algebra, *Physics Letters B* **785** (2018), 84–89. doi:10.1016/j.physletb.2018.08.032.
- [36] N. Furey, An algebraic roadmap of particle theories. Part II: theoretical checkpoints, *Annalen der Physik* **537** (2025), 2400323. doi:10.1002/andp.202400323.
- [37] N. G. Gresnigt, Three fermion generations and an untripllicated gauge sector with S_3 family symmetry, *Physics Letters B* **880** (2026), 140782. doi:10.1016/j.physletb.2026.140782.
- [38] J. C. Baez, Three generations in E_7 , arXiv:2608.06271 [hep-th], 2026.
- [39] J. M. Maldacena, The large N limit of superconformal field theories and supergravity, *Advances in Theoretical and Mathematical Physics* **2** (1998), 231–252. doi:10.4310/ATMP.1998.v2.n2.a1.
- [40] E. Witten, Anti de Sitter space and holography, *Advances in Theoretical and Mathematical Physics* **2** (1998), 253–290. doi:10.4310/ATMP.1998.v2.n2.a2.
- [41] R. Schiffler, *Quiver Representations*, CMS Books in Mathematics, Springer, Cham, 2014.
- [42] K. Kuwae and T. Shioya, Convergence of spectral structures: a functional analytic theory and its applications to spectral geometry, *Communications in Analysis and Geometry* **11** (2003), 599–673. doi:10.4310/CAG.2003.v11.n4.a1.
- [43] R. Howe, Remarks on classical invariant theory, *Transactions of the American Mathematical Society* **313** (1989), no. 2, 539–570. doi:10.1090/S0002-9947-1989-0986027-X.

A Tetrahedral representation bookkeeping

This appendix records the representation-theoretic decomposition used in Theorem 4.1. It is independent of the probabilistic origin of the source Gram.

Let $X = \{1, 2, 3, 4\}$ and let $\Phi = \{(i, j) : i \neq j\}$ be the twelve oriented edges. Root reversal is

$$\iota(i, j) = (j, i).$$

Let $P_{\mathbb{C}} = \mathbb{C}^{\Phi}$. Its even and odd reversal subspaces are

$$P^{\pm} = \{f : f(j, i) = \pm f(i, j)\}, \quad \dim P^{\pm} = 6.$$

A.1 The even sector

The even sector is the permutation representation on the six unordered edges of a tetrahedron. Let χ_+ be its character. For the five conjugacy classes of S_4 represented by

$$1, \quad (12), \quad (12)(34), \quad (123), \quad (1234),$$

the number of fixed unordered edges is

$$6, \quad 2, \quad 2, \quad 0, \quad 0.$$

Comparing with the S_4 character table gives

$$\boxed{P^+ \cong \mathbf{1} \oplus W \oplus V_2}, \quad (105)$$

where W is the standard three-dimensional representation and V_2 is the two-dimensional irreducible representation.

A.2 The odd sector

The odd sector is the oriented edge space $\Lambda^2 \mathbb{C}^4$ after identifying $e_i \wedge e_j$ with the signed edge $i \rightarrow j$. Write

$$\mathbb{C}^4 = \mathbb{C}u_0 \oplus W, \quad u_0 = \frac{1}{2}(1, 1, 1, 1).$$

Then

$$\Lambda^2 \mathbb{C}^4 = u_0 \wedge W \oplus \Lambda^2 W.$$

The first summand is isomorphic to W . Since $\dim W = 3$,

$$\Lambda^2 W \cong W \otimes \text{sgn}.$$

Therefore

$$\boxed{P^- \cong W \oplus (W \otimes \text{sgn})}. \quad (106)$$

The boundary map sends the first summand isomorphically onto the centred endpoint space and annihilates the second. Hence the two odd triplets are canonically distinguished as endpoint and cycle sectors.

Combining (105) and (106), removing the even root mean, and adjoining the scalar type contrast gives

$$\mathbb{C}_{\text{type}}^+ \oplus W_+ \oplus (V_2)_+ \oplus W_- \oplus (W \otimes \text{sgn})_-,$$

which is (48).

B Finite double-commutant and support-polar details

B.1 Bicommutant closure

For completeness, if $\mathcal{B} \subseteq \mathcal{B}(\mathcal{H})$ is a unital finite-dimensional C^* -algebra, then

$$\mathcal{B} \cong \bigoplus_{\alpha} M_{n_{\alpha}}(\mathbb{C}) \otimes I_{m_{\alpha}}$$

after a unitary decomposition of $\mathcal{H}$. Its commutant is

$$\mathcal{B}' \cong \bigoplus_{\alpha} I_{n_{\alpha}} \otimes M_{m_{\alpha}}(\mathbb{C}),$$

and therefore

$$\mathcal{B}'' = \mathcal{B}.$$

This is the only bicommutant input used in Theorem 3.11.

B.2 The support-polar algebra is intrinsic

The proof of Theorem 3.6 uses polynomial functional calculus only because the cutoff is finite. If

$$F^*F = \sum_{\lambda \in \Sigma} \lambda P_{\lambda}, \quad \Sigma \subset [0, \infty)$$

has finite spectrum, choose a polynomial g with

$$g(0) = 0, \quad g(\lambda) = \lambda^{-1/2} \quad (\lambda \in \Sigma \setminus \{0\}).$$

Then

$$U = Fg(F^*F)$$

is exactly the polar partial isometry. Hence neither support projection nor partial isometry is an additional finite input. The construction fails only if one asks for a uniformly bounded inverse as singular values collapse cofinally; the finite duality itself remains exact.

B.3 Basis independence of the quiver

Let

$$T_{ba} = \sum_{\alpha} E_{\alpha} \otimes B_{\alpha} = \sum_{\beta} \tilde{E}_{\beta} \otimes \tilde{B}_{\beta}$$

be two Hilbert–Schmidt orthonormal expansions of the same operator. The two coefficient families are related by a unitary transformation. Therefore their linear spans in $\mathcal{B}(N_a, N_b)$ agree. The arrow space $\mathfrak{B}_{ba}(T)$ is thus basis independent. The endomorphism algebra of the quiver is consequently canonical up to the unitary gauges of the multiplicity spaces and permutation of isomorphic central blocks.

C Multiplicity-birth and internal-router details

This appendix records the finite word-algebra facts behind the internal selection hierarchy of Section 4.1.2. The statements are independent of a particular stochastic presentation once the labelled source-word Gram and the external representation on its quotient are given.

C.1 Flat word depth is terminal

Let $\mathcal{L}$ be the finite represented source-letter alphabet on a source-minimal quotient and let

$$V_r = \text{span}\{\ell_k \cdots \ell_1 \xi : 0 \leq k \leq r, \ell_j \in \mathcal{L}, \xi \in V_0\}.$$

Let K_r be the Gram matrix of a labelled spanning family for V_r .

Lemma C.1 (Flat extension of a finite word span). *If $\text{rank } K_r = \text{rank } K_{r+1}$, then $V_{r+1} = V_r$, every letter in $\mathcal{L}$ preserves V_r , and $V_s = V_r$ for all $s \geq r$. Consequently every external isotypic multiplicity is constant for $s \geq r$.*

Proof. The inclusion $V_r \subseteq V_{r+1}$ is automatic. Equality of Gram ranks makes the two finite-dimensional quotient spans equal. Since every vector ℓv with $v \in V_r$ occurs among the words of length at most $r+1$, one has $\ell V_r \subseteq V_{r+1} = V_r$ for each letter ℓ . Induction on word length proves the remaining assertions. $\square$

Corollary C.2 (Terminating multiplicity obstruction). *At a flat word depth, failure of the multiplicity inequality $m_{\lambda,r} \geq n$ is a permanent obstruction to an internal $M_n(\mathbb{C})$ factor carried by the λ -multiplicity space.*

Proof. Apply Lemma C.1 and Theorem 4.2. $\square$

C.2 Router failure loci and generic saturation

For the diagonal algebra $\mathcal{D}_n$ of Theorem 4.5, define the router failure set

$$\mathfrak{F}_n = \{u \in M_n(\mathbb{C}) : C^*(\mathcal{D}_n, u) \neq M_n(\mathbb{C})\}.$$

For every nonempty proper subset $S \subset \{1, \dots, n\}$ let

$$L_S = \{u : p_i u p_j = p_j u p_i = 0 \text{ for all } i \in S, j \notin S\}.$$

Proposition C.3 (Algebraic structure of the router failure set). *One has*

$$\mathfrak{F}_n = \bigcup_{\emptyset \neq S \subsetneq \{1, \dots, n\}} L_S. \quad (107)$$

Each L_S is a proper complex linear subspace of $M_n(\mathbb{C})$. Hence $\mathfrak{F}_n$ is a proper Zariski-closed set and its complement is Zariski open, Euclidean open, and dense.

Proof. The support graph is disconnected exactly when some nontrivial union S of connected components has no support edge to its complement, which is precisely membership in L_S . Each L_S is defined by homogeneous linear equations on matrix entries and is proper because, for example, a cyclic shift matrix has connected support. A finite union of closed algebraic subsets is closed, and a finite union of proper linear subspaces cannot fill $M_n(\mathbb{C})$. $\square$

Proposition C.4 (Minimal bridge and weak-router residuals). *On $T \oplus H_{\text{priv}}$ define*

$$\delta_{\text{col}}(u) := \|p_H u p_T\|_{\text{HS}}^2 + \|p_T u p_H\|_{\text{HS}}^2.$$

On a weak multiplicity plane define

$$\delta_{\text{wk}}(p, q) := \|[p, q]\|_{\text{HS}}^2.$$

Then $\delta_{\text{col}} = 0$ exactly on the reducing colour branch $M_2 \oplus \mathbb{C}$, while $\delta_{\text{col}} > 0$ gives M_3 . For rank-one projections p, q on $\mathbb{C}^2$, one has

$$\delta_{\text{wk}}(p, q) = 2c^2(1 - c^2), \quad c = |\langle t, h \rangle|, \quad (108)$$

so $\delta_{\text{wk}} > 0$ exactly when the projections are distinct and nonorthogonal, in which case they generate M_2 .

Proof. The colour statement is Theorem 4.8. For the weak statement, choose a basis with $p = |e_1\rangle\langle e_1|$ and $h = ce_1 + \sqrt{1 - c^2}e_2$ after absorbing a phase. Direct multiplication gives (108). Its strict positivity is equivalent to $0 < c < 1$, which is the Burnside condition of Lemma 4.7. $\square$

C.3 A terminating internal landing certificate

The preceding results give a finite decision tree which does not fit an internal algebra by dimension. Start from a labelled physical word Gram and its external isotypic census.

Theorem C.5 (Finite internal landing alternative). *At any finite word depth exactly one of the following outcomes applies to the colour/weak landing problem.*

- (A1) *If the word hierarchy is flat and $m_{0,r} < 3$, an external-trivial internal colour $M_3(\mathbb{C})$ is impossible at every later depth.*
- (A2) *If $m_{0,r} \geq 3$ but every represented type-private bridge has $\delta_{\text{col}} = 0$, the colour block is reducible and remains $M_2 \oplus \mathbb{C}$ on the displayed three-space.*
- (A3) *If the weak multiplicity plane is present but every available pair has $\delta_{\text{wk}} = 0$, the represented weak multiplicity algebra is commutative.*
- (A4) *If the multiplicity census is $(3, 2, 1, 1)$ and both a positive colour residual and a positive weak-router residual occur, the represented internal word algebra equals the full relative commutant $M_3 \oplus M_2 \oplus \mathbb{C} \oplus \mathbb{C}$.*

On the fixed census $(3, 2, 1, 1)$ the intermediate algebra dimensions are exactly those of Theorem 4.13. Therefore every failure returns a finite algebraic witness and no failed condition is silently completed to the terminal branch.

Proof. Item (A1) is Corollary C.2. Item (A2) is Theorem 4.8; item (A3) is Lemma 4.7; and item (A4) is Theorem 4.12. The dimension statement is Theorem 4.13. $\square$

Remark C.6 (What this finite alternative does not claim). The theorem classifies and terminates the internal landing once the labelled source-word hierarchy is supplied. It does not assert that an arbitrary microscopic renewal law must realize the multiplicity birth, bridge, or router rows. Their occurrence is an operator-valued provenance question. This is precisely why the determinant line and matter incidence are tested only after the relative commutant has landed.

D Determinant incidence and anomaly bookkeeping

D.1 The determinant line

For $C \cong \mathbb{C}^3$ and $W_2 \cong \mathbb{C}^2$, define

$$\mathcal{D} = (\det C \otimes \det W_2)^*, \quad \mathcal{U} = \text{Hom}(C \otimes W_2 \otimes W_2, \Lambda^2 C^*).$$

For $\tau \in \mathcal{D}$, define

$$(\Theta_\tau(c \otimes w_1 \otimes w_2))(c_1, c_2) = \tau(c \wedge c_1 \wedge c_2 \otimes w_1 \wedge w_2).$$

Complete alternation in the three colour and two weak arguments gives a map $\mathfrak{A}_{\det} : \mathcal{U} \rightarrow \mathcal{D}$.

Proposition D.1 (Determinant-incidence line). *The maps Θ and $\mathfrak{A}_{\det}$ restrict to inverse isomorphisms between $\mathcal{D}$ and the fully alternating incidence line. In oriented orthonormal bases,*

$$\mathfrak{A}_{\det}(\Theta_\tau) = \tau, \quad \|\Theta_\tau\|_{\text{HS}}^2 = 6\|\tau\|^2.$$

Thus a nonzero fully alternating incidence shadow is equivalent to a nonzero determinant seed.

Proof. Complete alternation is the orthogonal projection onto the unique $(\det C \det W_2)^{-1}$ symmetry type. Evaluation on oriented orthonormal bases gives the factor $3!2!/2 = 6$ in the Hilbert–Schmidt normalization. $\square$

D.2 Hypercharge in left-handed convention

With $(a, b) = (-2, 3)$, the physical right-handed convention gives

$$\begin{aligned} Q &: (\mathbf{3}, \mathbf{2})_1, & u &: (\mathbf{3}, \mathbf{1})_4, & d &: (\mathbf{3}, \mathbf{1})_{-2}, \\ L &: (\mathbf{1}, \mathbf{2})_{-3}, & e &: (\mathbf{1}, \mathbf{1})_{-6}, & H &: (\mathbf{1}, \mathbf{2})_3. \end{aligned}$$

Writing all fermions as left-handed Weyl fields replaces

$$u \mapsto u^c : (\bar{\mathbf{3}}, \mathbf{1})_{-4}, \quad d \mapsto d^c : (\bar{\mathbf{3}}, \mathbf{1})_2, \quad e \mapsto e^c : (\mathbf{1}, \mathbf{1})_6.$$

The common kernel of these representations of $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)_y$ is generated by

$$\left(e^{2\pi i/3} I_3, -I_2, e^{i\pi/3} \right),$$

hence is $\mathbb{Z}_6$. This agrees with the quotient in (70).

E Finite common-action representative

The finite commutant duality is independent of a variational action. This appendix records one explicit local gauge–Higgs–Dirac representative on the reconstructed spacetime complex, together with the Ward, BRST, stress, and vacuum identities used only in the compatibility discussion of Section 4.3.

Let V_X, E_X, F_X be the finite spacetime vertices, oriented edges, and plaquettes of a reconstructed regulator, with positive Hodge weights $\star_{0,X}, \star_{1,X}, \star_{2,X}$. A finite internal configuration is

$$U_e \in G_{\text{SM}}, \quad H_v \in W_2, \quad \Psi_v \in \mathcal{S}_v \otimes G_{\text{gen}} \otimes \mathcal{H}_F^{(1)},$$

together with an independent dual amplitude $\bar{\Psi}_v$. Let $F_{p,X}$ be the principal-logarithm plaquette curvature on its local chart and let $D_U H$ be the covariant edge difference.

For positive gauge coefficients g_3, g_2, g_1 , scalar coefficient $\lambda_H > 0$, and $v_H \geq 0$, define

$$\begin{aligned} \mathcal{S}_{\text{SM},X}^E = & \frac{1}{2} \langle F_X, \star_{2,X} F_X \rangle_{g_3, g_2, g_1} + \frac{1}{2} \langle D_U H, \star_{1,X} D_U H \rangle \\ & + \sum_v m_{v,X} \lambda_H (\|H_v\|^2 - v_H^2)^2 \\ & + \text{Re} \langle \bar{\Psi}, \star_{0,X} (D_{\text{sp},X}^U + \Gamma_{\text{Cl}} \otimes D_F(H)) \Psi \rangle. \end{aligned} \tag{109}$$

The coefficients in (109) are physical parameters and are not fixed by Theorem 4.30.

Theorem E.1 (Finite Ward, BRST, and stress identities). *The action (109) is exactly gauge invariant and has cutoff-independent local support radius. Its Euler covectors obey the finite Ward identity*

$$\boxed{d_U^* E_U + \mathcal{R}_H^* E_H + \mathcal{R}_\Psi^* E_\Psi + \mathcal{R}_{\bar{\Psi}}^* E_{\bar{\Psi}} = 0.} \quad (110)$$

The standard ghost extension

$$sc = -c^2, \quad sU = c_t U - U c_s, \quad sH = cH, \quad s\Psi = c\Psi, \quad s\bar{c} = B, \quad sB = 0$$

is nilpotent and satisfies

$$\boxed{s^2 = 0, \quad s\mathcal{S}_{\text{SM},X}^E = 0.} \quad (111)$$

Metric/coframe variation defines the matter stress tensor, and common relabeling invariance gives

$$\boxed{\mathcal{L}_q^* \mathbf{T} = 2\mathcal{L}_U^* E_U + 2\mathcal{L}_H^* E_H + 2\mathcal{L}_\Psi^* E_\Psi + 2\mathcal{L}_{\bar{\Psi}}^* E_{\bar{\Psi}}.} \quad (112)$$

Proof. Plaquette holonomy transforms by conjugation, the Higgs edge difference covariantly in the target fibre, and the finite Dirac/Yukawa term intertwines the left and right typed representations. These identities prove gauge invariance. Differentiating gauge invariance gives (110). Nilpotence follows from the graded Jacobi identity and covariance. Differentiating common relabeling invariance gives (112). $\square$

Corollary E.2 (Stationary internal vacuum). *The configuration*

$$U_e = I, \quad H_v = v_H h_0, \quad \Psi_v = \bar{\Psi}_v = 0$$

is a stationary internal vacuum. For $v_H > 0$ its gauge stabilizer is

$$U(3) \cong \frac{\text{SU}(3) \times \text{U}(1)_{\text{em}}}{\mathbb{Z}_3}.$$

F Perturbative stability of the commutant gap

This appendix supplies the elementary estimate behind Theorem 4.42.

Let

$$\partial_{\mathbf{c}} : \mathcal{B}_2(\mathcal{H}) \rightarrow \mathcal{B}_2(\mathcal{H})^{\oplus s}, \quad \partial_{\mathbf{c}} X = ([c_1, X], \dots, [c_s, X]).$$

Then

$$\mathcal{L}_{\mathbf{c}} = \partial_{\mathbf{c}}^* \partial_{\mathbf{c}}.$$

If

$$\varepsilon = \left(\sum_j \|c_j - d_j\|_{\text{op}}^2 \right)^{1/2}, \quad C_{\mathbf{c}} = \left(\sum_j \|c_j\|_{\text{op}}^2 \right)^{1/2},$$

then

$$\|\partial_{\mathbf{c}} - \partial_{\mathbf{d}}\|_{2 \rightarrow 2} \leq 2\varepsilon, \quad \|\partial_{\mathbf{c}}\|_{2 \rightarrow 2} \leq 2C_{\mathbf{c}}.$$

Therefore

$$\begin{aligned} \|\mathcal{L}_{\mathbf{c}} - \mathcal{L}_{\mathbf{d}}\| &= \|\partial_{\mathbf{c}}^* \partial_{\mathbf{c}} - \partial_{\mathbf{d}}^* \partial_{\mathbf{d}}\| \\ &\leq \|\partial_{\mathbf{c}}\| \|\partial_{\mathbf{c}} - \partial_{\mathbf{d}}\| + \|\partial_{\mathbf{c}} - \partial_{\mathbf{d}}\| \|\partial_{\mathbf{d}}\| \\ &\leq 4(C_{\mathbf{c}} + C_{\mathbf{d}})\varepsilon. \end{aligned}$$

This proves the perturbative lower bound used in (90).

The perturbative estimate above is local in generator norm. The remainder of this appendix proves the compact-Mosco mechanism used in Section 4.2.5.

F.1 Transported forms and resolvent minimizers

Retain the embeddings $J_X : \mathcal{H}_X \rightarrow \mathcal{H}$ and the closed nonnegative forms q_X of Definitions 4.43 and 4.44. For $f \in \mathcal{H}$ let

$$u_X = (\mathcal{L}_X + I)^{-1} J_X^* f. \quad (113)$$

Equivalently, u_X is the unique minimizer on $D(q_X)$ of

$$\mathcal{F}_{X,f}(u) = q_X(u) + \|u\|^2 - 2 \operatorname{Re} \langle J_X u, f \rangle_{\mathcal{H}}. \quad (114)$$

Writing $q_X(\cdot, \cdot)$ for the sesquilinear form obtained by polarizing q_X , the Euler identity is

$$q_X(u_X, v) + \langle u_X, v \rangle = \langle f, J_X v \rangle_{\mathcal{H}}, \quad v \in D(q_X). \quad (115)$$

Taking $v = u_X$ gives

$$\|u_X\|^2 + q_X(u_X) \leq \|f\| \|u_X\| \leq \|f\|^2, \quad \|u_X\| \leq \|f\|. \quad (116)$$

Lemma F.1 (Mosco convergence gives strong transported resolvents). *Under Definition 4.43, for every fixed $f \in \mathcal{H}$,*

$$J_X(\mathcal{L}_X + I)^{-1} J_X^* f \longrightarrow J_\infty(\mathcal{L}_\infty + I)^{-1} J_\infty^* f \quad (117)$$

strongly in $\mathcal{H}$.

Proof. The estimate (116) makes $(J_X u_X)$ bounded. Every weakly convergent subsequence has, by the Mosco liminf inequality, a limit $J_\infty u$ satisfying

$$q_\infty(u) + \|u\|^2 - 2 \operatorname{Re} \langle J_\infty u, f \rangle \leq \liminf_X \mathcal{F}_{X,f}(u_X).$$

For every $v \in D(q_\infty)$ choose a Mosco recovery sequence v_X . Minimality of u_X gives $\mathcal{F}_{X,f}(u_X) \leq \mathcal{F}_{X,f}(v_X)$; passing to the limit shows that u minimizes $\mathcal{F}_{\infty,f}$. Strict convexity identifies $u = (\mathcal{L}_\infty + I)^{-1} J_\infty^* f$ and therefore identifies every weak cluster point. Comparing the minimum values and using the liminf inequality gives convergence of $\|u_X\|^2 + q_X(u_X)$ to the limiting value. Weak convergence together with convergence of the Hilbert norms yields strong convergence of $J_X u_X$. Thus the whole sequence converges. $\square$

Lemma F.2 (Asymptotic compactness gives collective compactness). *Assume Definition 4.44. If (f_X) is bounded in $\mathcal{H}$, then every sequence*

$$J_X(\mathcal{L}_X + I)^{-1} J_X^* f_X$$

has a strongly convergent subsequence. Hence the transported resolvents (R_X) form a collectively compact family. The limiting resolvent R_∞ is compact.

Proof. Apply (116) with $f = f_X$. The corresponding resolvent vectors have uniformly bounded norms and form energies, so asymptotic compactness gives a strongly convergent subsequence. The same argument for the limiting form, applied to an arbitrary bounded sequence of right-hand sides, proves compactness of R_∞ . $\square$

Lemma F.3 (Collectively compact self-adjoint convergence is norm convergence). *Let T_X be bounded self-adjoint operators on a Hilbert space. If $T_X \rightarrow T$ strongly and (T_X) is collectively compact, then T is compact and*

$$\|T_X - T\|_{\text{op}} \longrightarrow 0. \quad (118)$$

Proof. First let (x_n) be bounded. For each n choose a cofinal index X_n so large that $\|(T_{X_n} - T)x_n\| < 1/n$. Collective compactness gives a convergent subsequence of $T_{X_n}x_n$, and hence of Tx_n ; thus T is compact.

If (118) failed, there would be unit vectors x_X and $\varepsilon > 0$ with $\|(T_X - T)x_X\| \geq \varepsilon$. Passing to a subsequence, take $x_X \rightarrow x$, $T_X x_X \rightarrow y$ by collective compactness, and $Tx_X \rightarrow Tx$ by compactness of T . For every fixed z , self-adjointness and strong convergence give

$$\langle y, z \rangle = \lim_X \langle x_X, T_X z \rangle = \langle x, Tz \rangle = \langle Tx, z \rangle.$$

Thus $y = Tx$, so $(T_X - T)x_X \rightarrow 0$, a contradiction. $\square$

Proposition F.4 (Proof of compact Mosco resolvent convergence). *Under the hypotheses of Proposition 4.45,*

$$\|R_X - R_\infty\|_{\text{op}} \rightarrow 0.$$

Proof. Lemma F.1 gives strong convergence, Lemma F.2 gives collective compactness, and Lemma F.3 upgrades the convergence to operator norm. $\square$

F.2 Eigenvalues, Riesz projections, and kernel locking

Let $\mu_{1,X} \geq \mu_{2,X} \geq \dots > 0$ denote the nonzero eigenvalues of the compact positive operator R_X , repeated with multiplicity. On the screened space they are related to the eigenvalues of $\mathcal{L}_X$ by

$$\mu_{k,X} = \frac{1}{1 + \lambda_{k,X}}. \quad (119)$$

Lemma F.5 (Low-spectrum convergence). *If $\|R_X - R_\infty\| \rightarrow 0$, then for every fixed k ,*

$$\mu_{k,X} \rightarrow \mu_{k,\infty}, \quad \lambda_{k,X} \rightarrow \lambda_{k,\infty} \quad (120)$$

whenever the limiting eigenvalue is finite. If Σ is an isolated finite spectral cluster of R_∞ and Γ is a positively oriented contour separating it from the remaining spectrum, then the transported Riesz projections

$$\Pi_X(\Gamma) = \frac{1}{2\pi i} \int_\Gamma (z - R_X)^{-1} dz \quad (121)$$

converge to $\Pi_\infty(\Gamma)$ in operator norm. Their ranks are eventually equal.

Proof. For compact self-adjoint operators the min-max principle gives $|\mu_{k,X} - \mu_{k,\infty}| \leq \|R_X - R_\infty\|$ for each fixed k . Equation (119) gives the second convergence. For the projection statement, norm convergence and the resolvent identity imply uniform convergence of $(z - R_X)^{-1}$ on Γ ; integrate around the contour. If two projections differ in operator norm by less than one, their ranges have the same finite dimension, proving eventual rank stability. $\square$

Proposition F.6 (Protected-kernel locking). *Suppose each $\mathcal{M}_X$ is an m -dimensional subspace of $\text{Ker } \mathcal{L}_X$ and $\text{Ker } \mathcal{L}_\infty = \mathcal{M}_\infty$ has dimension m . Under compact Mosco convergence, for all sufficiently late X ,*

$$\text{Ker } \mathcal{L}_X = \mathcal{M}_X,$$

the transported kernel projections converge in operator norm, and

$$\lambda_{m+1}(\mathcal{L}_X) \rightarrow \lambda_{m+1}(\mathcal{L}_\infty) > 0.$$

Proof. The eigenvalue 0 of $\mathcal{L}_\infty$ corresponds to the eigenvalue 1 of R_∞ , of multiplicity m . Compactness of the limiting resolvent and finite-dimensionality of the kernel separate this eigenvalue from the rest of the spectrum. Lemma F.5 therefore gives a nearby spectral cluster of rank exactly m for all sufficiently late X . Because $\mathcal{M}_X \subseteq \text{Ker } \mathcal{L}_X$ already supplies m exact copies of the eigenvalue 1 of R_X , that entire cluster is the kernel and no extra zero mode can occur. The same lemma gives projection and next-eigenvalue convergence. $\square$

This proposition is the spectral step used in Theorem 4.46. It shows explicitly that the fixed protected zero modes and compact-Mosco convergence exclude spectral pollution at zero; norm-resolvent convergence and kernel stability are conclusions of the form hypotheses rather than additional assumptions.

G A one-generation incidence audit

The commutant language also gives a compact check of how Yukawa incidence reduces residual scalar symmetries inside one generation.

Let

$$\mathcal{F}_{15} = Q \oplus u \oplus d \oplus L \oplus e$$

with the modules of (74). Ignoring generation multiplicity for the moment, the gauge-generated algebra is

$$\mathcal{A}_{\text{gauge}} = M_6(\mathbb{C}) \oplus M_3(\mathbb{C}) \oplus M_3(\mathbb{C}) \oplus M_2(\mathbb{C}) \oplus \mathbb{C}, \quad (122)$$

because the five modules are irreducible and pairwise inequivalent. Hence

$$\mathcal{A}'_{\text{gauge}} \cong \mathbb{C}^5.$$

Proposition G.1 (Incidence reduces the one-generation commutant). *Assume the three Higgs-contracted incidence histories*

$$Q \leftrightarrow u, \quad Q \leftrightarrow d, \quad L \leftrightarrow e$$

are nonzero. Then

$$\boxed{\mathcal{A}_{\text{gauge+inc}} = M_{12}(\mathbb{C}) \oplus M_3(\mathbb{C}), \quad \mathcal{A}'_{\text{gauge+inc}} \cong \mathbb{C}^2.}$$

The two residual central directions are the quark and lepton connected components. Adding a neutral singlet and a nonzero L -neutral incidence changes the second matrix block to $M_4(\mathbb{C})$ while leaving the residual commutant $\mathbb{C}^2$.

Proof. A nonzero block between two full matrix summands, together with multiplication by endpoint matrix units, generates every rank-one map between the summands. Products and adjoints therefore generate the full matrix algebra on each connected component of the incidence graph. The three displayed edges have components $\{Q, u, d\}$ and $\{L, e\}$, of dimensions 12 and 3. $\square$

Remark G.2. This audit is intentionally distinct from the generation multiplicity theorem. It describes the commutant inside one fixed generation packet. The rank-three generation carrier of Proposition 4.27 is an additional multiplicity factor reconstructed before numerical flavour breaking is specified.

H Incidence determinant support and deletion margins

The rank theorem in Section 3.6 has a determinant refinement useful for finite audits. Keep the notation there and introduce nonnegative incidence weights $w = (w_e)_{e \in E}$:

$$G(w) = \sum_{e \in E} w_e G_e, \quad P_E(w) = \det G(w). \quad (123)$$

Theorem H.1 (Positive incidence determinant and sufficient supports). *The polynomial P_E is homogeneous of degree n and has nonnegative real coefficients. For any $S \subseteq E$, if $w_e > 0$ on S and $w_e = 0$ off S , then*

$$\boxed{P_E(w) > 0 \iff f(S) = n \iff \mathcal{M}(S) = \mathcal{M}(E)}. \quad (124)$$

If $\kappa_e = n - f(E \setminus \{e\})$, then the smallest power of w_e occurring with nonzero coefficient in P_E is exactly κ_e . Thus e is essential if and only if w_e divides P_E .

Proof. Stack the row blocks $\sqrt{w_e} D_e$. Cauchy–Binet expresses $\det G(w)$ as a sum of squared absolute values of $n \times n$ row minors, each multiplied by the monomial recording how many rows were selected from each edge block. All coefficients are therefore nonnegative. A point on a coordinate face has positive determinant exactly when the row spaces of the retained blocks span V , which is $f(S) = n$. For the last assertion, any row basis uses at most $f(E \setminus \{e\})$ rows from the other blocks and hence at least κ_e rows from block e ; extending a basis of the other-block span using rows from e attains this number. $\square$

For quantitative deletion, put $G = G_E \succ 0$ and

$$B_e = G^{-1/2} G_e G^{-1/2}, \quad \sum_e B_e = I, \quad B_R = \sum_{e \in R} B_e.$$

Proposition H.2 (Exact simultaneous-deletion margin). *For every $R \subseteq E$,*

$$G_{E \setminus R} = G^{1/2} (I - B_R) G^{1/2}, \quad (125)$$

and therefore

$$\boxed{\mathcal{M}(E \setminus R) = \mathcal{M}(E) \iff \|B_R\|_{\text{op}} < 1.} \quad (126)$$

If $\beta_R = \|B_R\|_{\text{op}} < 1$, then

$$G_{E \setminus R} \succeq (1 - \beta_R) G, \quad \lambda_{\min}(G_{E \setminus R}) \geq (1 - \beta_R) \lambda_{\min}(G).$$

Proof. The factorization follows from the definitions. Since $0 \preceq B_R \preceq I$, $I - B_R$ is positive definite exactly when its largest eigenvalue is smaller than one. Congruence by $G^{1/2}$ preserves nullity and positivity, giving both the commutant criterion and the lower bound. $\square$

Example H.3 (Minimal incidence skeletons need not form matroid bases). *On $\mathbb{C}^3$ with scalar base algebra let*

$$Y_a = E_{12} + E_{23}, \quad Y_b = E_{12}, \quad Y_c = E_{23}.$$

The singleton $\{a\}$ and the pair $\{b, c\}$ both generate $M_3(\mathbb{C})$ after adjoining adjoints, and hence both have scalar commutant. Each is inclusion-minimal: neither Y_b nor Y_c alone has scalar commutant. Thus minimal sufficient incidence sets can have unequal cardinality, so the incidence labels do not carry a matroid-basis structure in general.

I Historical operation, reciprocal-return, and determinant-source details

I.1 Large invariant word multiplicity need not be physical colour

Let $\Omega = \{(i, j) : 1 \leq i, j \leq 4, i \neq j\}$ and $\mathcal{H} = \mathbb{C}^\Omega$, with the simultaneous relabeling action $u_g|i, j\rangle = |g(i), g(j)\rangle$ of S_4 . Adjoin all diagonal point projections and the two fibre averages

$$(Tf)(i, j) = \frac{1}{3} \sum_{k \neq i} f(i, k), \quad (Hf)(i, j) = \frac{1}{3} \sum_{k \neq j} f(k, j).$$

Their support graph is connected, so the generated system algebra is $M_{12}(\mathbb{C})$.

Proposition I.1 (Invariant word-space inflation without an internal M_3). *For the preceding benchmark,*

$$\mathcal{A} = M_{12}(\mathbb{C}), \quad \mathcal{A}^{S_4} \cong \mathbb{C} \oplus M_2(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbb{C}.$$

In the irreducible order $(\mathbf{1}, \text{sgn}, W, W \otimes \text{sgn}, V_2)$, the multiplicities of the physical S_4 action on $\mathcal{H}$ are $(1, 0, 2, 1, 1)$, whereas the multiplicities of the conjugation action on $L^2(M_{12})$ are $(7, 5, 19, 17, 12)$. Thus the latter commutant contains an $M_7(\mathbb{C})$ block although the word-induced invariant system algebra has no $M_3(\mathbb{C})$ corner.

Proof. The permutation character on ordered unequal pairs is $(12, 2, 0, 0, 0)$ on the conjugacy classes $1, (12), (12)(34), (123), (1234)$. Character inner products with the five irreducibles of S_4 give $(1, 0, 2, 1, 1)$ and hence the displayed physical commutant. Conjugation on $M_{12} \cong \mathcal{H} \otimes \overline{\mathcal{H}}$ has character $|\chi_{\mathcal{H}}|^2 = (144, 4, 0, 0, 0)$, whose character inner products are $(7, 5, 19, 17, 12)$. Theorem 2.8 retains only left multiplication by $\mathcal{A}^{S_4}$ as word-induced invariant system actions. $\square$

I.2 Subdirect products and central separation

Theorem I.2 (Finite subdirect matrix-algebra classification). *Let $\mathcal{B} \subseteq \bigoplus_{i=1}^r M_{n_i}(\mathbb{C})$ be a unital C^* -subalgebra whose coordinate projection onto every displayed factor is surjective. Then the coordinates partition into classes of equal matrix size, and $\mathcal{B}$ contains one independent full matrix algebra per class, embedded diagonally across that class up to inner automorphisms. In particular, distinct matrix sizes cannot be diagonally identified.*

Proof. Write the intrinsic decomposition $\mathcal{B} \cong \bigoplus_{\beta} M_{m_{\beta}}(\mathbb{C})$. A surjective $*$ -homomorphism to a simple factor $M_{n_i}(\mathbb{C})$ annihilates every intrinsic summand except one: the images of the intrinsic central units are orthogonal central projections summing to the identity, so exactly one is nonzero. On that summand the map is a nonzero homomorphism between simple full matrix algebras and hence an isomorphism, forcing $m_{\beta} = n_i$. Coordinates seeing the same intrinsic summand are related by automorphisms of a full complex matrix algebra, hence by inner automorphisms. Different intrinsic summands are independent. $\square$

Theorem 4.11 is the specialization to block sizes $(3, 2, 1, 1)$. It explains why independent scalar supports, rather than blockwise surjectivity alone, are needed before the four dimensions in Theorem 4.13 may be added.

I.3 A route-bank stopping projection

For the mediator notation of Theorem 4.9, set

$$\mathcal{N}_A = \text{span}\{h^k \text{Ran } \rho_A : 0 \leq k < n\}, \quad \mathcal{N}_B = \text{span}\{h^k \text{Ran } \rho_B : 0 \leq k < n\}.$$

Cayley–Hamilton makes both spaces h -invariant, and self-adjointness of h makes them reducing.

Proposition I.3 (All-excursion return alternative). *If $\mathbf{m}_{\text{ret}} = 0$, then $\mathcal{N}_A \perp \mathcal{N}_B$ and*

$$Q = I_T \oplus 0_{H_{\text{priv}}} \oplus (I_V \otimes P_{\mathcal{N}_A})$$

commutes with every generator of the declared reciprocal route bank and its adjoint. Consequently every word w in that bank has $p_{Hwp_T} = 0$. Thus failure of the finite return panel is an exact obstruction to arbitrarily long repeated excursions within the same bank.

Proof. The equality $\mathbf{m}_{\text{ret}} = d^{-1} \text{Tr}(\rho_B W_A) = 0$ and positivity imply $\text{Ran } \rho_B \perp \mathcal{N}_A$, hence $\mathcal{N}_B \perp \mathcal{N}_A$. Partial-trace positivity gives $\text{Ran } A \subseteq V \otimes \text{Ran } \rho_A$ and $\text{Ran } B \subseteq V \otimes \text{Ran } \rho_B$. The stated projection therefore reduces the entry, exit, mediator and adjoint generators, contains the type line and annihilates the private plane. Every generated word has zero $H_{\text{priv}} \leftarrow T$ corner. $\square$

I.4 Sharp arity of the determinant character

Let a native finite carrier have defining central weights $(1, 0)$ on $C \cong \mathbb{C}^3$, $(0, 1)$ on $W_2 \cong \mathbb{C}^2$, and $(0, 0)$ on any neutral sector. The centre of $U(3) \times U(2)$ acts on a weight (a, b) by $e^{i(a\alpha + b\beta)}$.

Theorem I.4 (Sharp five-slot determinant-mode floor). *The central-weight support of the single-system endomorphism algebra contains only*

$$(0, 0), \quad \pm(1, 0), \quad \pm(0, 1), \quad \pm(1, -1),$$

and therefore does not contain the determinant-incidence weight $(-3, -2)$. On a diagonal k -fold tensor carrier this character is absent for $k < 5$ and is present for $k = 5$.

Proof. Single-system endomorphism weights are target-minus-source differences among $(1, 0), (0, 1), (0, 0)$, giving the displayed list. In k tensor slots every vector weight (a, b) has $a, b \geq 0$ and $a + b \leq k$, so a target-minus-source weight with total negative charge -5 cannot occur for $k < 5$. At $k = 5$ the alternating line in $C^{\otimes 3} \otimes W_2^{\otimes 2}$ transforms by $\det C \det W_2$; a rank-one map from that line to a neutral line therefore has weight $(-3, -2)$. $\square$

This result concerns tensor arity, not temporal word length. It explains why longer evolution inside a fixed low-arity memory cannot substitute for the typed five-leg incidence of Proposition 4.17.

J Source identifiability and finite system tomography

This appendix gives a finite witness for Theorem 2.4 and an explicit panel realizing Corollary 2.6. The construction is algebraic and is not assigned particle-physics semantics.

Let H_E be a finite edge-system space, let Q_a ($1 \leq a \leq 4$) be nonzero orthogonal projections on a record factor H_R with $\sum_a Q_a = I$, and let $K_1, \dots, K_6 \in \mathcal{B}(H_E)$ satisfy

$$\sum_{e=1}^6 K_e^* K_e = I, \quad \sum_{e=1}^6 K_e = \sqrt{2} I, \quad K_1, \dots, K_6 \text{ linearly independent.} \quad (127)$$

For $0 < \vartheta < 1$ set

$$A_{e,a} = K_e \otimes Q_a, \quad L_0 = \sqrt{1-\vartheta} I, \quad L_{e,a} = \sqrt{\vartheta} A_{e,a}.$$

These operators define a normalized resolved instrument.

Choose a real 6×5 matrix O with

$$O^\top O = I_5, \quad O^\top \mathbf{1} = 0, \quad O O^\top = I_6 - \frac{1}{6} \mathbf{1} \mathbf{1}^\top. \quad (128)$$

Proposition J.1 (Five-Kraus freedom behind the six-edge normalization). *Equation (127) holds if and only if there are five matrices $B_1, \dots, B_5$ with $\sum_\mu B_\mu^* B_\mu = I$ and $I, B_1, \dots, B_5$ linearly independent such that*

$$K_e = \frac{1}{\sqrt{18}} I + \sqrt{\frac{2}{3}} \sum_{\mu=1}^5 O_{e\mu} B_\mu. \quad (129)$$

Consequently the coarse edge channel has the form

$$\mathcal{E}_* = \frac{1}{3} \text{id} + \frac{2}{3} \Psi_*, \quad \Psi_*(\rho) = \sum_\mu B_\mu \rho B_\mu^*,$$

and $C^*(K_e : e) = C^*(B_\mu : \mu)$.

Proof. Apply the orthogonal coefficient change whose first column is $\mathbf{1}/\sqrt{6}$ and whose remaining columns are O . The coherent-sum constraint fixes the first transformed coefficient to $I/\sqrt{3}$; the other five, $D_\mu = \sum_e O_{e\mu} K_e$, satisfy $\sum_\mu D_\mu^* D_\mu = 2I/3$. Put $B_\mu = \sqrt{3/2} D_\mu$ and invert the orthogonal change. $\square$

J.1 An exact nonidentifiability pair

Let $\mathcal{S}$ be the twenty three-element subsets of $\{1, \dots, 6\}$. On $\mathbb{C}^{\mathcal{S}}$ define commuting Hermitian involutions

$$H_e^c |S\rangle = (2\mathbf{1}_{e \in S} - 1) |S\rangle.$$

Let $\Gamma_1, \dots, \Gamma_5$ be five anticommuting Hermitian Clifford matrices on $\mathbb{C}^4$, and on $\mathbb{C}^4 \otimes \mathbb{C}^5$ set

$$H_e^q = \sqrt{\frac{6}{5}} \sum_{\mu=1}^5 O_{e\mu} \Gamma_\mu \otimes I_5.$$

For $b \in \{c, q\}$ put

$$K_e^b = \frac{1}{\sqrt{18}} I_{20} + \frac{i}{3} H_e^b. \quad (130)$$

Proposition J.2 (Record-complete nonidentifiability witness). *Both lists in (130) satisfy (127), and*

$$(K_e^b)^* K_e^b = I_{20}/6, \quad \text{Tr}[(K_e^b)^* K_f^b] = \begin{cases} 10/3, & e = f, \\ 2/3, & e \neq f. \end{cases} \quad (131)$$

Yet

$$C^*(K_e^c) = \mathbb{C}^{20}, \quad C^*(K_e^q) = M_4(\mathbb{C}) \otimes I_5. \quad (132)$$

For the resolved instrument above, every uninterrupted word in the original classical alphabet has the same effect in the two realizations. A word with $m \geq 1$ accepted letters, n_0 no-response letters and one common accepted record type a has effect

$$(1 - \vartheta)^{n_0} (\vartheta/6)^m I_{20} \otimes Q_a; \quad (133)$$

if accepted record types disagree the effect is zero. Moreover the two one-step opportunity channels have the same nonzero Choi spectrum,

$$\text{spec}_+(J_{\text{opp}}) = \left\{ \left[\frac{20}{3} (12 - 11\vartheta) \right]^{(1)}, \left(\frac{20\vartheta}{3} \right)^{(3)}, \left(\frac{8\vartheta}{3} \right)^{(20)} \right\}. \quad (134)$$

Thus the complete classical record channel at every word length is identical although the system algebras differ.

Proof. For the commuting family, counting three-subsets gives zero mean and pair correlation $-1/5$. For the Clifford family the row vectors of $\sqrt{6/5}O$ are unit vectors with the same pair correlations, and Clifford anticommutation gives $(H_e^q)^2 = I$. Hence both families satisfy the same normalization and Gram relations, and each $\sqrt{6}K_e^b$ is unitary. The twenty sign patterns distinguish all basis vectors in the commuting realization, giving $\mathbb{C}^{20}$. The five Clifford generators span and generate $M_4(\mathbb{C}) \otimes I_5$. Finally, every classical word effect is $L_w^* L_w$; because each accepted system factor is $6^{-1/2}$ times a unitary and the record projectors are orthogonal, this gives (133) in both realizations.

For the one-step Choi spectrum, the Gram of the twenty-four accepted Choi vectors is $G_{24} = G_6 \otimes I_4$ with $G_6 = (8/3)I_6 + (2/3)\mathbf{1}\mathbf{1}^\top$. It has eigenvalue $20/3$ on the uniform edge direction with four record directions and $8/3$ on the twenty edge-contrast directions. The coefficient matrix $M_\vartheta = \vartheta I + (1 - \vartheta)cc^*$ acts by $12 - 11\vartheta$ on the global uniform direction and by ϑ on its orthogonal complement. The nonzero Choi eigenvalues are therefore those of $M_\vartheta^{1/2} G_{24} M_\vartheta^{1/2}$, giving (134). $\square$

J.2 Ten coherent pullbacks reconstruct the six coefficients

Enumerate the twenty-four accepted operators $A_{e,a}$ as A_j and let B synthesize their Choi vectors $|A_j\rangle\rangle$. In $\mathbb{C}^{24}$ put

$$c = \mathbf{1}/\sqrt{2}, \quad M_\vartheta = \vartheta I + (1 - \vartheta)cc^*.$$

Let C_ν be the matrices synthesized by the columns of $B M_\vartheta^{1/2}$ and define the minimal Stinespring isometry

$$V_\vartheta = \sum_{\nu=1}^{24} C_\nu \otimes |\nu\rangle, \quad \mathcal{P}_{\vartheta,*}(\rho) = \text{Tr}_{\text{sys}}(V_\vartheta \rho V_\vartheta^*).$$

For $z \in \mathbb{C}^{24}$ write $C(z) = (I \otimes \langle z|) V_\vartheta$. Then $\mathcal{P}_\vartheta^*(|z\rangle\langle w|) = C(z)^* C(w)$.

For the twenty-four coefficient labels (e, a) define

$$u_{e,a} = 1/\sqrt{24}, \quad (r_\mu)_{e,a} = O_{e\mu}/2, \quad \langle u, r_\mu \rangle = 0,$$

and the fixed Hermitian environment quadratures

$$X_\mu = |u\rangle\langle r_\mu| + |r_\mu\rangle\langle u|, \quad Y_\mu = -i|u\rangle\langle r_\mu| + i|r_\mu\rangle\langle u|. \quad (135)$$

Let $\mathcal{P}_\vartheta^*$ denote the Heisenberg complementary pullback of the normalized coefficient frame and write $X_\mu = \mathcal{P}_\vartheta^*(X_\mu)$, $Y_\mu = \mathcal{P}_\vartheta^*(Y_\mu)$. Set $h = 12 - 11\vartheta$ and $D_\mu = \sum_e O_{e\mu} K_e$.

Proposition J.3 (Explicit ten-panel reconstruction). *The ten Hermitian operator-valued pullbacks determine all six system coefficients through*

$$D_\mu \otimes I_{H_R} = \frac{2\sqrt{3}}{\sqrt{\vartheta h}}(X_\mu + iY_\mu), \quad K_e = \frac{I}{\sqrt{18}} + \sum_\mu O_{e\mu} D_\mu. \quad (136)$$

Thus this finite operator-valued export determines the represented system algebra and commutant.

Proof. In the normalized coefficient frame the uniform environment vector is an eigenvector of the coefficient Gram with eigenvalue h , while each r_μ has eigenvalue ϑ . Hence $C(u) = \sqrt{h/12} I$ and $C(r_\mu) = \sqrt{\vartheta} D_\mu \otimes I_{H_R}/2$. Since $X_\mu + iY_\mu = 2|u\rangle\langle r_\mu|$, Equation (7) gives the first reconstruction formula; orthogonality of O gives the second. $\square$

Proposition J.4 (Robust commutant stopping test). *If the ten pullbacks have Hilbert–Schmidt errors $\epsilon_{X,\mu}$ and $\epsilon_{Y,\mu}$, then*

$$\left(\sum_e \|\hat{K}_e - K_e\|_{\text{HS}}^2 \right)^{1/2} \leq e_K := \frac{2\sqrt{3}}{\sqrt{\vartheta h}} \left[\sum_\mu (\epsilon_{X,\mu} + \epsilon_{Y,\mu})^2 \right]^{1/2}. \quad (137)$$

For the star-closed stacked commutator map,

$$\|\hat{\mathcal{D}} - \mathcal{D}\|_{2 \rightarrow 2} \leq 2\sqrt{2} e_K. \quad (138)$$

Consequently a measured least singular value $\hat{s} > 2\sqrt{2} e_K$ on an independently verified protected complement certifies absence of any further commutant direction there.

Proof. Equation (136) and the triangle inequality give the D_μ errors. Orthogonality of O preserves the sum of squared coefficient errors. For unit Hilbert–Schmidt X , each coefficient error changes a commutator by at most twice its operator norm; retaining both a coefficient and its adjoint gives the factor $2\sqrt{2}$. The last statement is the standard least-singular-value perturbation inequality. $\square$

Remark J.5. The ten-panel theorem is a sufficient finite tomography statement, not an assertion that these coherent quadratures are automatically present in every microscopic source. If only the classical resolved record is supplied, Proposition J.2 shows that the acted-on system algebra can remain underdetermined. Conversely, once the operator-valued pullbacks are physically available on a common frame, the reconstruction is exact and feeds directly into the commutant audit of the main text.